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A Comprehensive Review of Deep Transfer Learning for Intelligent Fault Diagnosis in Rotating Machines

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ABSTRACT

Rotating machinery serves as a critical backbone for national economic growth and is extensively utilized as mechanical equipment across diverse industrial domains. However, failures in these machines can cause significant operational disruptions, financial losses, and safety risks. This has led to an increased focus on advancing intelligent fault diagnosis techniques, with industrial big data and artificial intelligence (AI) emerging as powerful tools. Recently, deep transfer learning (DTL) has gained significant attention as a promising approach for cross-domain and cross-machine diagnosis, particularly in cases with limited faulty data and complex conditions. Despite significant progress, a comprehensive review of recent advancements in DTL remains absent, and well-defined future research directions for its development are lacking. To address this gap, this review paper provides an extensive analysis of the existing literature regarding the recent applications of DTL in the fault diagnosis of rotating machines. This review categorizes existing DTL methods for machine fault diagnosis into four key aspects: instance-based, feature-based, parameter-based, and adversarial-based transfer. The survey first introduces the basic concepts of DTL, then explores recent research applications in depth and provides valuable perspectives in the field. In addition, the review examines the strengths and weaknesses of different DTL types in the fault diagnosis of rotating machines. Building on recent research progress, this review outlines emerging challenges and potential future research directions in the field. Thus, the review article aims to serve as a valuable resource and inspire further exploration among researchers, policymakers, and industry experts in the domain.

1 | Introduction

With the rapid progression of science and technology, mechanical equipment in modern industry is evolving to become more functional and complex. Among this equipment, rotating machinery is particularly significant, utilizing rotary motion to perform various tasks and representing over 90% of modern industrial machinery [1]. Hence, this machinery plays a crucial role in industrial production and is extensively employed across diverse sectors, including aerospace, construction, transportation, energy, and more, with applications in turbines, fans, generators, centrifugal pumps, and other systems [2–4]. During prolonged operation under varying loads, these machines expose key components, including bearings and gears, to wear, fatigue, and fractures in complex and dynamic conditions. In severe cases, such failures can lead to equipment shutdowns, production halts, substantial financial losses, and major safety risks. Consequently, fault diagnosis in rotating machinery has become a core priority in system design and maintenance [5, 6]. Over time, extensive research and engineering efforts have led to the development of diverse techniques for rotating machine fault diagnosis. These approaches can be broadly classified into four categories based on their evolution: knowledge-based, physical model-based, signal-based, and data-driven methods [7, 8]. Among them, data-driven methods have gained prominence in machine fault diagnosis with the growth of modern industry, driven by advancements in sensors and computer systems [9].

More recently, the integration of AI and industrial intelligence has further enhanced data-driven fault diagnosis, leveraging industrial big data and intelligent analysis to provide reliable and robust solutions in the field. Owing to its fast computation and strong generalization, AI has gained significant attention from academia and industry, reducing reliance on domain-specific expertise. A wide range of AI techniques has proven effective in diverse fields such as face recognition [10] and healthcare [11]. Moreover, AI has demonstrated remarkable potential in addressing complex challenges, including identifying unknown dynamics in time-delayed nonlinear systems [12, 13] and enabling adaptive control of nonlinear dynamical systems [14]. Building on research advances, AI-driven fault diagnosis for rotating machines is generally categorized into three major approaches: conventional machine learning (ML), deep learning (DL), and transfer learning (TL). These categories, illustrated in Figure 1, encompass a range of methods, each with distinct techniques for detecting faults in rotating machines.

Conventional ML-based fault diagnosis for rotating machines follows three key stages: data acquisition, feature extraction, and fault classification. In data acquisition, sensors like vibration, temperature, current, and acoustic emission capture raw signals indicative of machine health. Feature extraction involves manually analyzing data attributes and selecting relevant features based on expert-defined criteria, which may introduce redundancy or irrelevant information [15, 16]. Finally, fault classification maps extracted features to specific fault types using methods such as random forests [17], support vector machines (SVMs) [18], k-nearest neighbors [19], decision trees [20], artificial neural networks (ANNs) [21], and Naïve Bayes [22]. Despite notable progress, these methods still struggle to meet rising industrial demands: shallow models and manual feature extraction limit their ability to process complex, high-dimensional data; separating feature extraction from decision-making hinders synchronized optimization and increases computation; and growing sensor diversity and machine complexity further challenge their effectiveness, resulting in suboptimal fault diagnosis [23].

DL, the most prominent branch of ML, has revolutionized fault diagnosis by enabling automatic feature extraction from raw data and boosting diagnostic accuracy [24–26]. This growth is driven by both internal factors, such as feature learning, advanced data processing, and architectural innovations, and external factors, including the explosion of industrial big data, hardware breakthroughs, and increasing industrial demands. Over the past

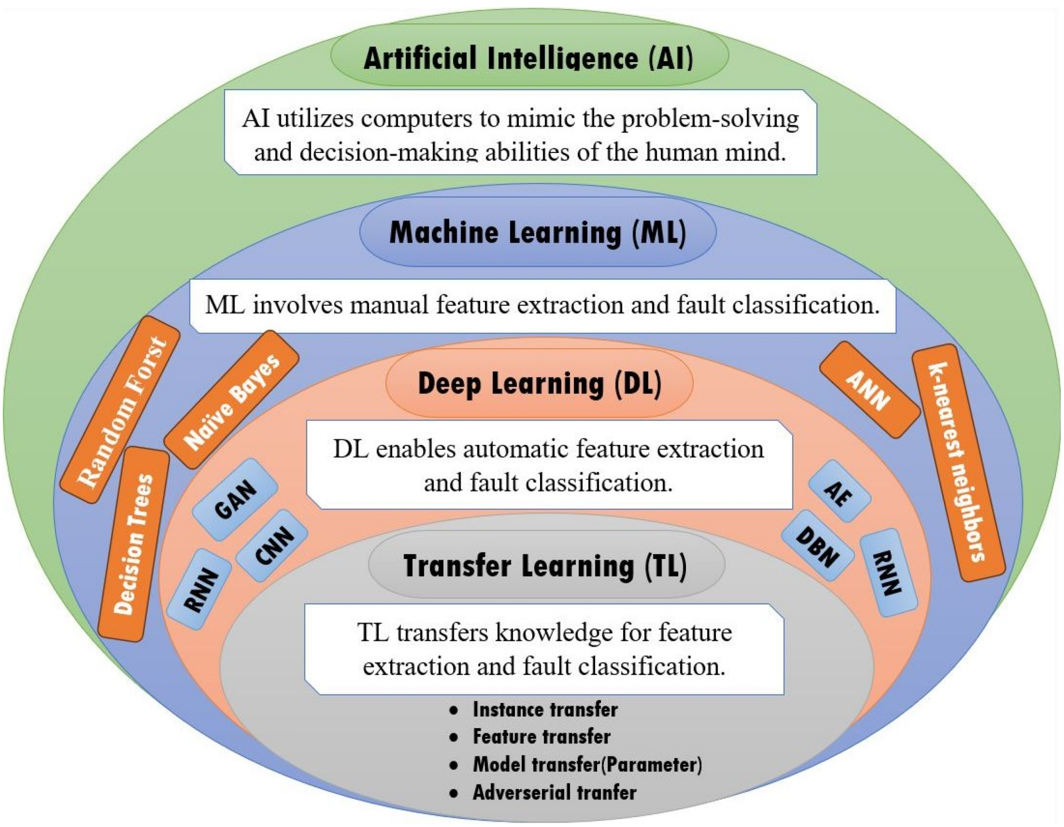

FIGURE 1 | Research advancements in AI-based fault diagnosis for rotating machines, representing ML, DL, and TL, each with distinct techniques.

decade, DL has propelled intelligent fault diagnosis in rotating machines, with widely used models including autoencoders (AEs) [27], convolutional neural networks (CNNs) [28], deep belief networks (DBNs) [29], generative adversarial networks (GANs) [30], and recurrent neural networks (RNNs) [31]. While significant advancements have been made in DL-based machine fault diagnosis, two critical challenges remain: the need for a sufficient number of labeled samples for effective model training and the requirement that training and testing samples be independently and identically distributed [32]. In practical engineering contexts, meeting both criteria simultaneously proves challenging. First, the large volumes of condition monitoring data require costly and time-consuming manual labeling. Second, variations in mechanical operations due to changes in load, speed, and environmental conditions create distribution shifts in vibration data. These issues hinder the development of accurate, generalizable DL models and limit the broader application of fault diagnosis in rotating machinery.

Recently, to address these two prevalent issues, TL has been increasingly applied to rotating machine fault diagnosis, leveraging existing knowledge across related domains. Deep transfer learning (DTL) further enhances this by integrating DL, enabling automatic fault feature extraction while mitigating data distribution discrepancies [6, 33]. As a result, diagnostic knowledge gained under one operational condition can be effectively transferred to new conditions, improving model generalization and robustness in industrial settings. Compared to traditional methods, DTL improves cross-task learning, automatically extracts expressive features, and supports end-to-end diagnosis. By transferring knowledge from related tasks and adapting to data variations, it enables fault diagnosis with minimal or no additional labeled data, avoiding model retraining and reducing reliance on labeled datasets.

Building on recent advancements, DTL-based fault diagnosis has gained significant momentum, leading to a surge in research publications. Given the growing adoption of DTL in rotating machine fault diagnosis, a comprehensive review is essential to synthesize existing work, identify the research challenges, and outline the promising future directions. Prior to our study, some review articles had already explored the application of DTL in this domain. For example, Li et al. [34] provided an overview of TL, focusing on its role in fault feature extraction and classification. In addition, Hakim et al. [35] specifically reviewed TL techniques for bearing fault diagnosis, while Yu and Zhang [36] examined its application in process fault detection and diagnosis. Similarly, Chen et al. [37] surveyed TL-driven intelligent fault diagnosis, offering insights and perspectives, while Yan et al. [33] presented a comprehensive review of DTL for anomaly detection in industrial time series. Furthermore, Yang et al. [38] reviewed mechanical fault diagnosis using DTL, and Guo et al. [6] analyzed adversarial-based DTL for mechanical fault diagnosis.

While existing reviews on TL in fault diagnosis offer valuable perspectives, most focus exclusively on either algorithmic advancements or practical applications. However, a dedicated emphasis on rotating industrial machine fault diagnosis, along with a comprehensive integration of DTL algorithms with real-world applications, is still lacking. Recognizing this gap underscores the significance and necessity of this study. In addition, previous reviews have summarized recent research progress but have largely overlooked emerging DTL theories, despite their widespread adoption in the field. Moreover, a clear roadmap for future advancements in DTL-based rotating machine fault diagnosis methods is often missing. To address these gaps, this paper presents a comprehensive and up-to-date review of the latest developments in the domain. Unlike prior studies, this work goes beyond summarizing research by critically analyzing the strengths, limitations, and variations of different DTL approaches. Furthermore, traditional and state-of-the-art DTL algorithms are systematically categorized, applications across diverse industrial scenarios are examined, and existing challenges and future research directions are highlighted. This study not only serves as a reference for researchers and practitioners but also provides valuable perspectives to drive the advancement of DTL technologies in rotating machine fault diagnosis.

Consequently, this review article categorizes existing DTL methods for fault diagnosis in rotating machines into four primary categories: instance-based transfer, model or parameter-based transfer, feature or mapping-based transfer, and adversarial-based transfer. Each category is systematically summarized, outlining the core concepts, typical models, and significant studies related to the field.

In summary, the main contributions of this paper are as follows:

1. The review paper provides an in-depth exploration of the fundamental concepts and various methods employed in DTL for rotating machine's fault diagnosis. This serves as a valuable resource for early-career researchers and industry professionals seeking a deeper understanding of the field and offers essential guidance for integrating DTL into intelligent fault diagnosis designs.

2. The paper summarizes the research progress and conducts a detailed analysis of the latest literature on DTL approaches, enhancing the understanding of state-of-the-art methods in rotating machine fault diagnosis. This overview ensures that researchers remain informed about the latest advancements and techniques in the domain.

3. The review systematically categorizes advancements in rotating machine fault diagnosis methods based on DTL into the four main groups mentioned earlier. For each DTL category, the paper outlines representative algorithms and summarizes their development and evolution, enabling readers, researchers, and practitioners to gain a coherent understanding of DTL variants for machine fault diagnosis and their research progress.

4. This paper critically examines the strengths and weaknesses of different types of DTL methods based on their advancements in the domain. This analysis significantly aids researchers in selecting appropriate DTL approaches tailored to specific improvements, rather than making random selections.

5. Furthermore, the paper identified potential research challenges and limitations while proposing perspectives for future work based on existing literature. Following these

directions can enhance fault diagnosis accuracy and reliability in rotating machines, contributing to optimized performance, reduced expenses, and enhanced risk mitigation in industries such as energy and manufacturing. Thus, the findings provide actionable recommendations for professionals, AI practitioners, researchers, and decision-makers involved in machine fault diagnosis and the health management of industrial machinery.

Going forward, Section 2 presents the research method for this review. Section 3 introduces the core principles of DTL and classifies its methods. Section 4 explores advancements and applications of DTL in rotating machine fault diagnosis. Section 5 discusses key findings, identifies major challenges in existing research, and outlines promising directions for future exploration. Finally, Section 6 concludes the review and outlines future prospects.

2 | Research Method

In this paper, the authors conduct a comprehensive review to identify, select, and critically evaluate research on intelligent fault diagnosis of rotating machines using DTL methods. The review process includes defining a clear search strategy, establishing databases, selecting precise keywords, setting inclusion criteria, and specifying the publication time frame. The literature search was performed using two prominent indexing databases: Scopus and Web of Science. Initially, a preselection list of papers aligned with the focused topic was compiled based on their titles. Subsequently, only publications that matched the topic according to their abstracts and full content screening were included. To identify DTL research in the field, articles were searched using keywords such as "fault diagnosis," "fault detection," "transfer learning," and "deep transfer learning" in titles, abstracts, or keywords. The focus then shifted to application-oriented studies leveraging DTL for diagnosing faults in rotating machines. In addition, a targeted literature search was conducted in Scopus using the following query string:

> ((fault diagnosis OR fault classification OR fault detection OR fault identification) AND (deep transfer learning OR DTL) AND (rotating machine OR rotating equipment OR gear OR compressor OR turbine OR motor OR bearing OR pump OR engine) AND (transfer learning OR TL))

This strategy was carefully designed to capture a broad spectrum of studies and maximize the retrieval of research on DTL in the context of rotating machinery diagnostics. Given the large volume of search results, it was not feasible to review every publication in detail; therefore, a rigorous screening process was implemented to identify the most relevant and high-quality studies on DTL for fault diagnosis in rotating machines. The selection process followed these inclusion criteria: (i) publication year between 2015 and 2025, with some earlier papers included to illustrate the development of methods; (ii) written in English; (iii) subject areas limited to Engineering or Computer Science; and (iv) article types including journal papers, conference papers, reviews, or book chapters. The exclusion criteria were: (i) source types such as trade journals or book series; (ii) articles still in press; and (iii) works published only as abstracts. To ensure quality and relevance, papers lacking sufficient information on DTL for intelligent fault diagnosis in rotating machines, as well as duplicates, were excluded. In addition, references of all included papers were manually checked to identify further relevant studies in the field. Ultimately, 184 papers meeting the established criteria were included, and their key contributions were synthesized to form the basis of this comprehensive review. The following section presents the main findings derived from this in-depth analysis.

3 | Theoretical Background of DTL

TL was first introduced as an algorithmic model by Pan and Yang in early 2009 [39]. This approach, inspired by the human learning process, enables the reuse of knowledge or skills acquired from different yet related tasks to address new challenges. The fundamental principle of TL is to identify common features between two or more related but non-identical learning tasks and apply this shared knowledge to improve learning efficiency. In doing so, TL reduces training time and minimizes the need for labeled data. This approach is particularly useful when labeled data in the target domain is limited [40].

In real-world fault diagnosis scenarios for rotating industrial machines such as gearboxes, pumps, turbines, and bearings, the availability of normal operational data significantly outweighs the amount of fault data. This imbalance arises because industry regulations typically prohibit prolonged operation of faulty machinery. As a result, the scarcity of fault data can severely impact the diagnostic effectiveness of DL models. In addition, variations in operating conditions such as fluctuations in load, speed, and environmental factors introduce inconsistencies in the collected vibration data, disrupting the assumption of a consistent data distribution. These challenges make it difficult for conventional DL methods to maintain robust performance [6, 41, 42]. To address these issues, TL plays a critical role in mitigating data scarcity and the assumption that training and testing samples must follow identical distributions. Hence, TL improves fault diagnosis accuracy by leveraging knowledge from a source domain to enhance model performance in a target domain, even under varying real-world conditions [32].

In TL-based rotating machine fault diagnosis, datasets are generally categorized into two domains: the source domain, representing previously learned conditions, and the target domain, reflecting new or unseen operational scenarios. To maintain alignment with prior surveys on rotating machine fault diagnosis, the datasets can be succinctly described using mathematical formulations, along with the key definitions and notations employed in TL. Moving forward, in TL, two primary concepts in TL are involved: the domain (D) and the task (T).

A domain D is defined by a feature space X and a marginal probability distribution $P(X)$. The feature space consists of samples:

$$X = \left\{ x_i | x_i \in X, i = 1, \ldots, n \right\} \tag{1}$$

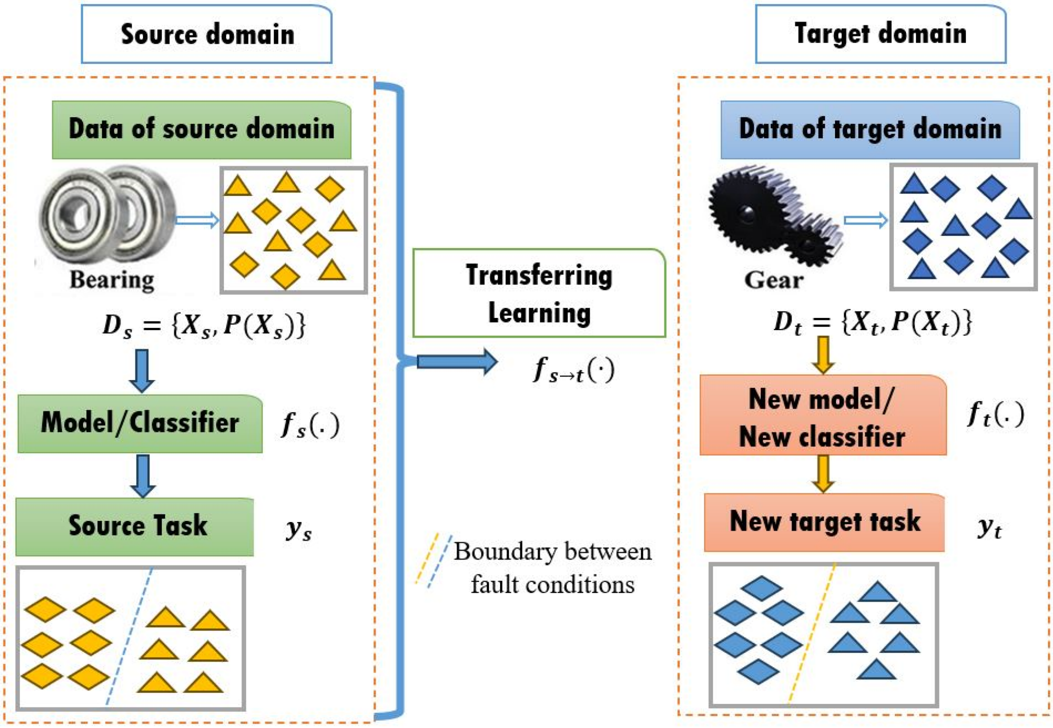

FIGURE 2 | Transfer learning workflow for fault diagnosis, with knowledge flow from source to target domain.

where n is the number of feature vectors and x_i represents the ith feature vector. For fault diagnosis, X corresponds to the space of fault features, and each x_i represents a sampled faulty feature vector.

For a given domain $D = \{X, P(X)\}$, a task T is represented as:

$$T = \{Y, f(\cdot)\} \tag{2}$$

where Y denotes the label space and $f(\cdot)$ is the predictive function that maps features to fault labels. The predictive function learns the relationship between feature vectors x_i and labels y_i:

$$\left\{\left(x_i y_i\right) | x_i \in X, y_i \in Y\right\} \tag{3}$$

From a probabilistic perspective, the predictive function can also be expressed as:

$$f(x) = P(y|x)$$

Then, in TL, datasets are divided into a source domain D_s and a target domain D_t, formally defined as:

$$D_s = \left\{x_s^i, y_s^i\right\}_{i=1}^{K_s} \tag{4}$$

$$D_t = \left\{x_t^i, y_t^i\right\}_{i=1}^{K_t} \tag{5}$$

where K_s and K_t denote the number of samples in the source and target domains, respectively. Here, x_s^i and x_t^i represent the ith feature vectors, while y_s^i and y_t^i represent their corresponding labels. Typically, $Ks \gg Kt \geq 0$.

The corresponding tasks in the source domain T_s and target domains T_t are defined as:

$$T_s = \left\{Y_s, f_s(\cdot)\right\} \tag{6}$$

$$T_t = \left\{Y_t, f_t(\cdot)\right\} \tag{7}$$

where Y_s and Y_t denote the label spaces, and $f_s(\cdot)$ and $f_t(\cdot)$ are the predictive functions for the source and target domains.

Finally, based on the above definitions and notations, TL for fault diagnosis: given a source domain D_s with its corresponding task T_s and a target domain D_t with task T_t, the goal is to learn a mapping function:

$$f_{s\to t}(\cdot) \tag{8}$$

This function minimizes the discrepancy between the source and target feature distributions while enhancing the predictive performance of the target task $f_t(\cdot)$, even when $D_s \neq D_t$ or $T_s \neq T_t$. For further detail, Figure 2 schematically depicts the transfer process in fault diagnosis and the flow of knowledge from the source domain to the target domain.

Based on the development of TL, Deep TL (DTL) is a rapidly advancing machine learning framework that integrates the powerful feature representation capabilities of DL with efficient knowledge adaptation. Its remarkable performance has led to widespread adoption across various domains, including image recognition [43], text processing [44], and speech processing [45], garnering significant attention from researchers. Moreover, DTL has demonstrated substantial success in diagnosing faults in rotating machines, offering a more robust, adaptable and reliable approach [46]. This has fueled extensive research and development in intelligent fault diagnosis, further enhancing its applicability in real-world industrial settings.

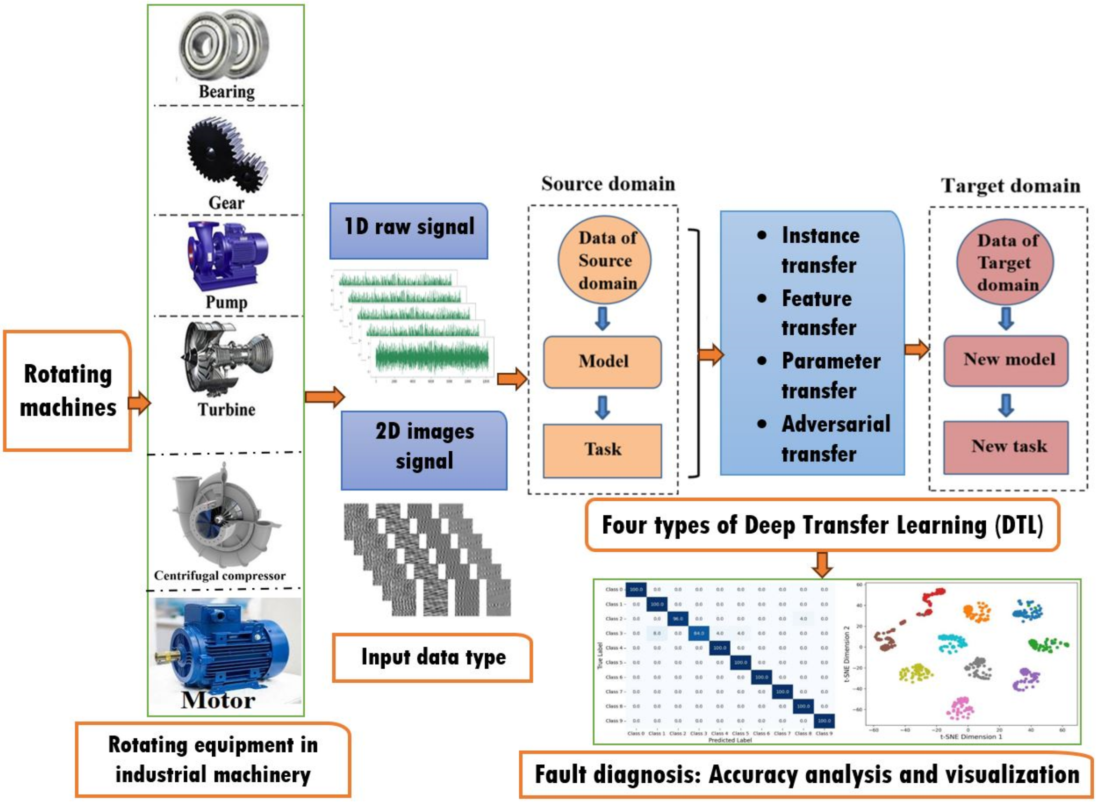

FIGURE 3 | General overview of the four DTL methods for fault diagnosis in rotating industrial machines, including input data types and the diagnostic procedure.

Building on recent research advancements, DTL-based fault diagnosis approaches for rotating machines are classified into four categories: feature/mapping-based, instance-based, parameter/pretrained model-based, and adversarial-based transfer methods [24]. Feature transfer, instance transfer, and domain-adversarial transfer are classified as data-driven approaches, concentrating on knowledge transfer by utilizing substantial amounts of data. These methods typically involve transforming and adjusting data instances or manipulating data from diverse domains through techniques such as feature alignment and feature mapping. Conversely, parameter transfer represents a model-driven approach that prioritizes understanding the fundamental structure and dynamics of the data. This process generally entails transferring the parameters of a pretrained model from the source domain to the target domain [33]. This classification reflects the significant progress and growing popularity of DTL techniques in the fault diagnosis of rotating machines. In the detail understanding of the four types of DTL, as shown in Figure 3, this classification encompasses rotating equipment fault diagnosis methods alongside presenting input data types (1D signals and 2D images). The classification also includes source–target domain transfer via four DTL types, and final fault diagnosis with accuracy evaluation and visualization.

Consequently, this review paper places particular emphasis on newly emerging DTL methods tailored for fault detection in rotating industrial machines. The next section delves into the latest developments in DTL methods and their practical applications in this domain.

4 | Improvements and Applications of DTL Methods for Fault Diagnosis of Rotating Machines

This section provides a comprehensive review of the most widely used DTL approaches and their respective categories: feature or mapping-based, instance-based, parameter or pretrained model-based, and adversarial-based methods applied to the fault diagnosis of rotating machines. The section elaborates on the principles behind each DTL category, the methods involved, and their applications within the field.

4.1 | Instance Transfer-Based Fault Diagnosis Methods for Rotating Machines

The instance-based DTL approach focuses on leveraging source domain data by reweighting instances. This method selects and prioritizes relevant data that closely aligns with the target domain. The key steps in this approach involve identifying useful instances from the source domain for the target task and assigning weights to these instances based on their relevance to the target domain [33, 47]. Building on advancements in instance-based knowledge transfer, this approach is applied when the source and target domains share similar feature spaces but exhibit differences in data distribution due to varying operational conditions or loads. In addition, it facilitates fault diagnosis in new machine models by utilizing data from older, comparable systems. Consequently, instance transfer-based fault diagnosis methods for rotating machines are developed to address the challenge of limited labeled samples in the target domain. These methods acknowledge that the available labeled data in the target domain is insufficient for constructing a robust diagnostic model. Therefore, the primary objective is to leverage relevant samples from the source domain to develop a more reliable diagnostic model, ultimately enhancing performance on the target domain's dataset [48–50].

In instance-based DTL, transfer adaptive boosting (TrAdaBoost) is a prominent method for fault diagnosis in the field. By combining TrAdaBoost with DTL principles, it enhances classification performance when labeled target-domain samples are limited. The approach leverages labeled data from a related source domain while iteratively adjusting the importance of source and target samples to reduce misclassification. The mathematical formulations, together with the key definitions and notations used in TrAdaBoost, are presented as follows:

The weight vector W assigns importance to each training sample during the learning process. It is defined as:

$$W = [w_1, w_2, \ldots, w_m] \tag{9}$$

where m represents the total number of training samples available from both the source and target domains, and w_i denotes the weight associated with the ith sample. Higher weights indicate that the sample is more important for the current iteration of training, while lower weights reduce the influence of that sample.

The learning rate in TrAdaBoost determines the extent to which sample weights are updated during each iteration. Mathematically, it is given by:

$$\beta = \frac{1}{1 + \frac{2}{\ln^2}} \tag{10}$$

This parameter controls how strongly the algorithm emphasizes misclassified samples while down weighting correctly classified ones. With a fixed learning rate, TrAdaBoost balances source and target samples in each boosting round, ensuring stable convergence. Here, T denotes the total number of iterations. In each round, a weak classifier h_t (a decision stump or simple neural network) is trained, and sample weights are updated based on its error. Finally, the predictions of all weak learners are combined to form a strong classifier.

The classification error at iteration t (e_t) quantifies the performance of the base classifier h_t on the weighted training data:

$$e_t = \sum_{i=1}^{m} W_i \cdot \text{sgn}\left(h_t(x_i) - y_i\right) \tag{11}$$

where y_i is the true label of sample x_i, and sgn($\cdot$) is the sign function, which returns +1 if the prediction is correct and −1 if it is incorrect (or vice versa, depending on the implementation).

A higher error means the classifier misclassified more samples, leading the algorithm to adjust sample weights more aggressively in later iterations. Train (W) denotes the training function that generates the base classifier h_t from the current weighted dataset. This function ensures that higher-weighted samples exert greater influence during training, enabling the algorithm to focus on misclassified or underrepresented target samples. Weighted training is central to TrAdaBoost ability to adaptively transfer knowledge from the source domain while enhancing target classification performance. Figure 4 illustrates the instance-based transfer learning process in TrAdaBoost, showing how knowledge is iteratively transferred through weight adjustments.

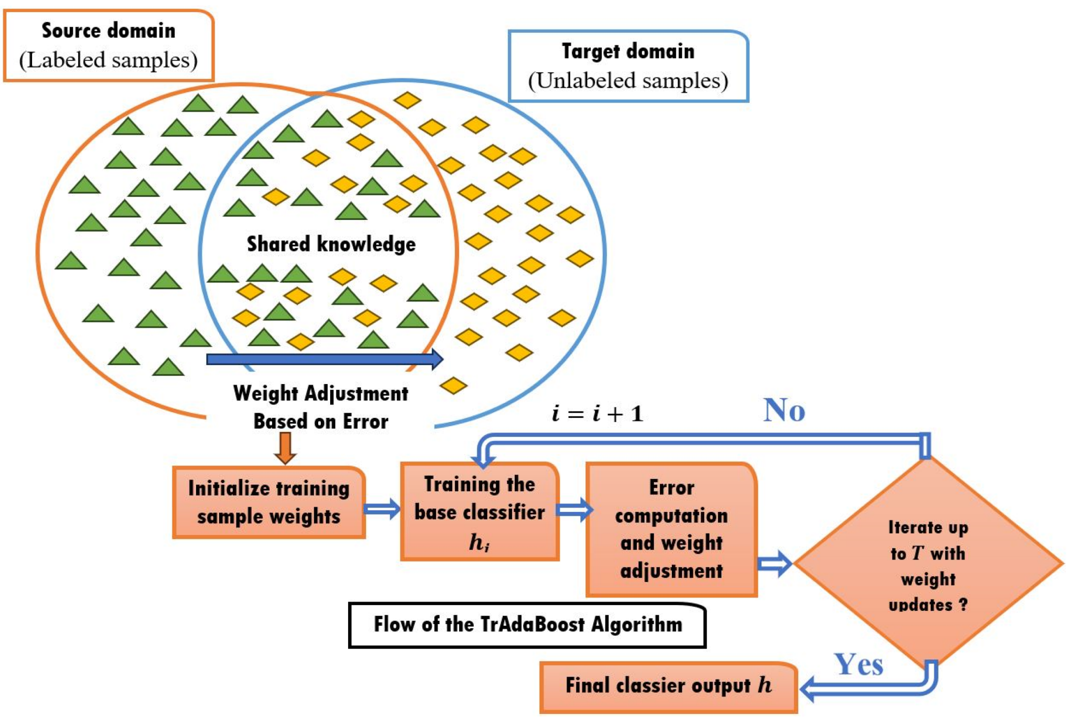

FIGURE 4 | Procedure of instance-based DTL for knowledge transfer, showing the steps of the TrAdaBoost algorithm and its iterative weight adjustment process.

During each iteration of the TrAdaBoost training process, if the diagnostic model misclassifies a source domain sample, its weight is reduced, indicating a significant difference from the target domain. Over time, the weights of source samples that positively impact target classification are increased, while those of less relevant samples are decreased. In addition, the weights of misclassified target samples are raised, consistent with the core concept of TrAdaBoost. Hence, leveraging these advancements, TrAdaBoost has been effectively utilized in rotating machine fault diagnosis. For instance, Xiao et al. [49] introduced a motor fault diagnosis model leveraging the TrAdaBoost algorithm with CNN as the base classifier, achieving high accuracy despite limited target domain data. Likewise, Dong et al. [51] applied TrAdaBoost-based TL to a marine turbocharger diagnosis model, while Du et al. [52] developed a hybrid TL approach using transfer component analysis and TrAdaBoost for machinery fault diagnosis under variable conditions. In addition, Miao et al. [50] addressed data scarcity in seawater hydraulic pump fault diagnosis using TrAdaBoost TL, while Chen et al. [53] demonstrated TrAdaBoost effectiveness in handling imbalanced and distributionally different wind turbine fault data. Moreover, Annaz et al. [54] developed a framework integrating uniform manifold approximation and projection for dimensionality reduction and transfer adaptive boosting for nuanced instance weighting and classification.

In addition, instance-based transfer methods employed in intelligent fault diagnosis include importance sampling/weighting, which probabilistically re-weights source domain samples according to their similarity to the target distribution. For example, Li et al. [55] applied instance weighting-based partial domain adaptation for fault diagnosis of rotating machinery, while Lee et al. [56] introduced a multi-objective instance weighting deep transfer learning network for enhanced intelligent fault diagnosis.

Moreover, instance-based DTL has been successfully applied to rotating machines as an alternative to TrAdaBoost and importance sampling/weighting methods. For example, Jamil et al. [57] proposed deep boosted TL to mitigate negative transfer in wind turbine gearbox fault detection, while Shi et al. [58] developed an instance-adaptive multisource transfer approach using dynamic feature encoding for fault diagnosis of rotating machinery. Similarly, Skowron [59] demonstrates the effectiveness of CNN-based TL for fault diagnosis in permanent magnet synchronous motors, utilizing an instance-based transfer approach. Furthermore, Zhang et al. [60] proposed a cross-domain decision method that uses weighted structure expansion and reduction with instance transfer to address disparities between source and target data distributions in fault diagnosis.

Overall, instance-based DTL effectively diagnoses faults in rotating machines by leveraging relevant instances from a similar source domain while adapting to different operational conditions. This approach enhances adaptability and reduces the need for extensive labeled data, making it efficient and cost-effective. In addition, the diagnostic approach is straightforward to comprehend and implement, serving as a robust tool when applied appropriately. However, the method often necessitates a minimal disparity between the source and target domains; otherwise, it can suffer from improper weight distribution and negative transfer. Additionally, poor instance selection can lead to overfitting and diminished performance. Challenges such as computational complexity and difficulties in diagnosing new fault types can further restrict its real-world applicability. Ultimately, the key challenge in instance-based transfer is selecting source samples with a similar distribution to the target domain for training a more accurate model.

4.2 | Feature Transfer-Based Fault Diagnosis Methods for Rotating Machines

When the source and target domains differ significantly, instance-based transfer learning often fails to extract directly transferable knowledge. In contrast, feature-based transfer learning addresses this challenge by assuming that, despite instance-level variations, both domains share underlying feature patterns. Domain-invariant representations are learned by mapping data into a common feature space, thereby extracting and aligning fault-related features, reducing discrepancies, and improving diagnostic accuracy. A central technique is domain adaptation (or mapping transfer), which learns transformation functions to project data from both domains into a shared space [24]. The goal is to minimize divergence in latent feature distributions and achieve relevant target-domain representations through feature alignment, mapping, and encoding [61]. In feature-based DTL for rotating machine fault diagnosis, several metrics are used to quantify distribution discrepancies between the source and target domains. These include Maximum Mean Discrepancy (MMD) [62], Kullback–Leibler (KL) Divergence [63], and Wasserstein distance [64], among others. Within these, MMD is the most commonly utilized metric for mapping transfer, as evidenced by its frequent application in the reviewed literature [33]. In feature-based or mapping transfer, MMD is used to measure the distance between the source domain D_s and the target domain D_t. This distance serves as a key criterion in domain-adaptive learning, and its formal definition is given as:

$$Dist\left(D_s, D_t\right) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \emptyset\left(d_{s_i}\right) - \frac{1}{n_t} \sum_{j=1}^{n_t} \emptyset\left(d_{t_i}\right) \right\|_H^2 \tag{12}$$

where n_s and n_t are the numbers of source and target samples, respectively, and $\phi(\cdot)$ represents the feature mapping function in a high-dimensional Reproducing Kernel Hilbert Space.

To facilitate computation, MMD employs the kernel trick by constructing the kernel matrix K and the matrix L with elements L_{ij}, defined as:

$$K = \begin{bmatrix} K_{S,S} & K_{S,T} \\ K_{T,S} & K_{T,T} \end{bmatrix}, L_{ij} = \begin{cases} \frac{1}{n_s^2}, & d_i, d_j \in D_s \\ \frac{1}{n_t^2}, & d_i, d_j \in D_t \\ -\frac{1}{n_s n_t}, & \text{Otherwise} \end{cases} \tag{13}$$

With this formulation, Equation (12) can be simplified into a more compact trace form:

$$Dist\left(D_s, D_t\right) = tr(KL) - \mu \cdot tr(K)$$

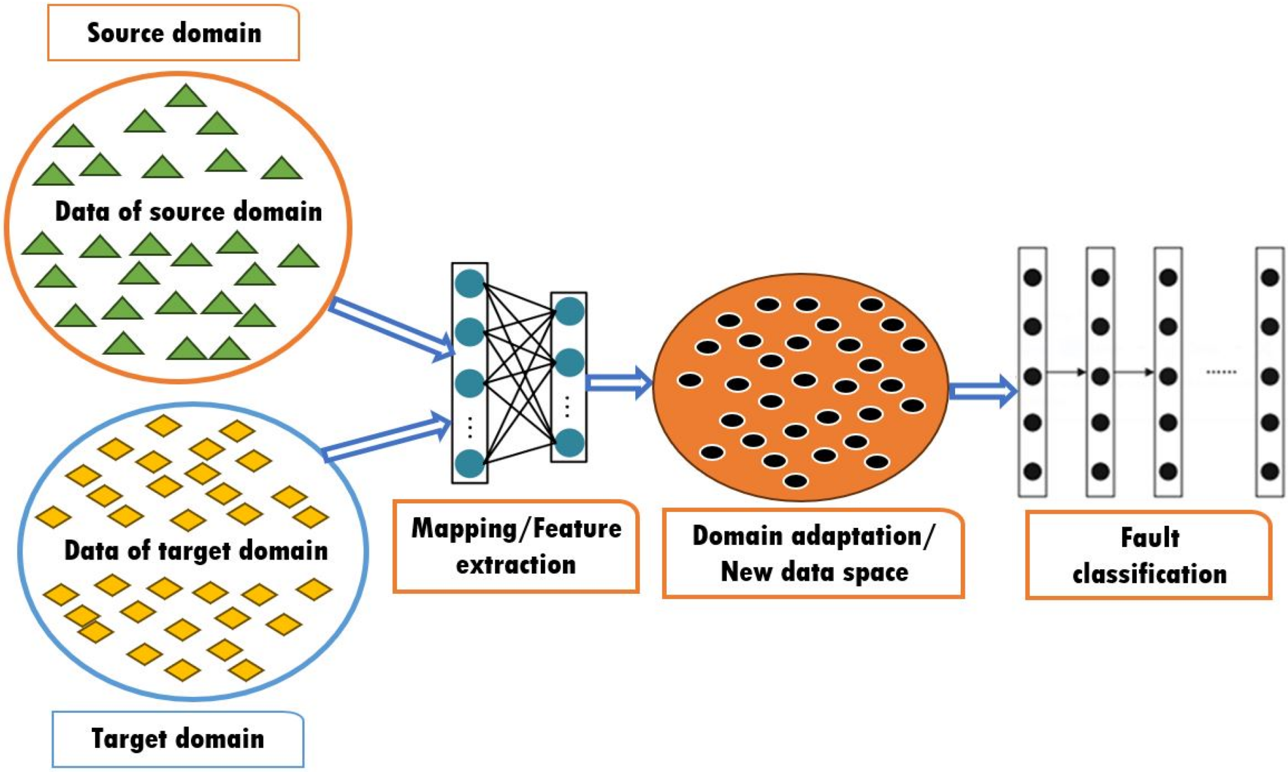

FIGURE 5 | Overview of feature-based transfer methods for aligning source and target feature distributions.

where μ is a balancing parameter that regulates the trade-off between aligning distributions and preserving the raw data structure.

In the framework of feature transfer, the optimization objective then becomes:

$$\min_{W} tr\left(W^T LKW\right) \cdot \mu \cdot tr\left(W^T W\right), s.t. W^T HKHW = I_m \tag{14}$$

where $H = I_{n_s - n_t} - \frac{1}{(n_s + n_t)} 11^T$ is the centering matrix.

As illustrated in Figure 5, minimizing the MMD criterion aligns source and target feature distributions. This alignment ensures that knowledge extracted from the source domain is more effectively transferred to the target domain, thereby enhancing model generalization and significantly improving fault diagnosis accuracy under varying conditions [65–67].

As a result, advancements in MMD metrics have significantly enhanced feature transfer for fault diagnosis in rotating machines. For instance, Zhang et al. [68] developed a CNN model for cross-domain fault diagnosis, leveraging MMD to minimize domain discrepancies. Similarly, Yang et al. [69] integrated CNN with MMD to design a TL framework for diagnosing faults in locomotive bearings. In addition, Li et al. [70] introduced a CNN-based approach for rolling bearing fault diagnosis, incorporating a feature clustering technique to enhance intra-class similarity, while Zhu et al. [71] proposed a DTL method utilizing multi-Gaussian kernel MMD for diagnosing rolling bearing faults. Moreover, Xiao et al. [72] introduced a domain adaptation strategy for motor fault diagnosis, using CNNs to extract hierarchical features and integrating MMD to reduce distribution disparities. Furthermore, Xu et al. [73] proposed a convolutional transfer discrimination network for unbalanced fault diagnosis, employing MMD to align high-dimensional feature distributions.

Building on the success of MMD in mapping-based TL for rotating machine fault diagnosis, Yu et al. [74] proposed an unsupervised adaptation framework with a multi-scale feature extractor and a multi-kernel MMD (MK-MMD)-based generator to align labeled and unlabeled industrial data. Similarly, Huang et al. [75] introduced a multi-source model for bearing diagnosis, employing MMD to select relevant sources and Wasserstein distance to align classifier outputs, reducing misclassification. Qian et al. [76] applied maximum mean square discrepancy to enhance domain confusion in wind turbine gearbox diagnosis, capturing mean and variance in reproducing kernel Hilbert space without requiring labeled target samples. Moreover, Li et al. [77] developed a robust DTL for cross-device bearing fault diagnosis, integrating multi-domain feature extraction with multi-attention and BiGRU fusion, while MK-MMD minimized domain discrepancies. Furthermore, Luo and Chen [78] used CNN with MK-MMD for gearbox diagnosis across varying speeds, and Thuan [79] proposed a robust TL method for bearing diagnosis using a shallow neural network with multilayer MMD loss to adapt source-to-target features.

In addition, ongoing research has expanded domain adaptation techniques, including Transfer Component Analysis (TCA) [80], Joint Distribution Adaptation (JDA) [81], Deep Adaptation Network (DAN) [82], and Adaptive Batch Normalization (AdaBN). TCA, a widely used feature-based DTL method, maps source and target domain data into a Hilbert space to align their marginal distributions, minimizing domain discrepancies. Its advancements have significantly improved fault diagnosis in rotating machines by reducing cross-domain distribution differences, extracting transferable fault features, and enabling ML-based fault identification. For instance, Xu et al. [66] utilized TCA to extract transferable features from raw vibration signals, followed by a k-NN classifier for diagnosing bearing faults. Similarly, Chen et al. [65] demonstrated the effectiveness of TCA in capturing cross-domain fault features in bearings. Building on this approach, Xu et al. [83] first applied wavelet packet decomposition to raw signals, then used TCA to align source and target domain feature sets, and performed fault identification in industrial bearings. In addition, to enhance diagnosis speed and accuracy, Li and Ma [84] integrated an enhanced TCA with a DBN, enabling subdomain adaptation and improving sample compactness. While these studies primarily addressed minor domain variations, van de Sand et al. [85] used a TCA-based model with adaptive SVM for fault diagnosis across heterogeneous chillers.

JDA extends TCA by aligning both marginal and conditional distributions of cross-domain data. For example, Xu et al. [86] enhanced rolling bearing diagnosis using JDA and adaptive filtering to handle chaotic time-domain features under varying conditions. Similarly, Tong et al. [67] extracted transferable features via JDA and used k-NN classification. To further improve TL effectiveness, Han et al. [87] integrated JDA with deep CNNs for fault diagnosis. Recently, to strengthen domain adaptation, Qin et al. [88] proposed an advanced JDA alignment strategy that addresses both marginal and conditional distribution discrepancies, enabling efficient fault diagnosis for wind turbine gearboxes and cross-bearing data. Likewise, Chang et al. [89] introduced the dynamical JDA network, a cross-machine fault diagnosis framework that uses dynamic weighting to optimize domain alignment and class discrimination. Xiong et al. [90] introduced a multi-kernel JDA with dynamic distribution alignment for bearing fault diagnosis, enhancing accuracy and cross-domain robustness. Additionally, Liu and Dong [91] proposed a supervised JDA framework for bearing fault diagnosis, combining empirical mode decomposition, feature extraction, and refined domain adaptation. Moreover, Xu et al. [92] advanced JDA by integrating soft clustering and graph-embedded alignment with subspace weighting and instance reweighting for effective source–target domain synchronization. Furthermore, Geng et al. [93] proposed a domain adaptation approach that integrates JDA with adversarial learning to address open-set challenges in intelligent bearing fault diagnosis.

On the other hand, DAN, which is more widely adopted than JDA, incorporates MMD distance into deep networks for domain adaptation, refining feature alignment during training, while MK-MMD further enhances adaptation. For instance, Wan et al. [94] applied sparse DAE with domain adaptation for gearbox diagnosis, outperforming traditional methods. To address label scarcity in new conditions, Che et al. [95] developed a domain-adaptive DBN trained via MK-MMD, validated on bearing fault diagnosis. Similarly, Xu et al. [96] designed a dynamic DAN for bearing fault diagnosis under variable conditions, integrating a stacked sparse autoencoder, dynamic distribution adaptation, and a domain-invariant classifier. Additionally, Lu et al. [97] proposed a semi-supervised contrastive DAN for rotating machinery fault diagnosis, integrating in-domain contrastive learning and local MMD to enhance feature adaptation.

Alternative adaptation strategies include AdaBN. For instance, Qian et al. [98] proposed a three-stage DTL model with AdaBN for rotating machine fault diagnosis, eliminating the need for target datasets. Likewise, Li et al. [99] developed a bearing diagnosis method leveraging AdaBN to improve generalization across diverse data distributions and environments, ensuring superior adaptability and real-time responsiveness. To further enhance diagnosis, Li et al. [100] introduced a TL-driven approach combining AdaBN with an optimization algorithm, using a ResNet trained on source data and adapted to the target domain to mitigate domain shift. Similarly, Reference [101] presented an enhanced AdaBN transfer learning method for bearing fault diagnosis by optimizing batch normalization in DL architectures. Beyond AdaBN, Cheng et al. [102] applied Wasserstein distance (WD) minimization to extract CNN-based features from raw vibration signals, enhancing domain alignment for high-accuracy fault diagnosis. In addition, Zhu et al. [103] introduced an improved WD-based TL method for fault diagnosis, learning transferable features between labeled and unlabeled signals from different equipment. Moreover, Lei et al. [104] developed a DTL-based fault diagnosis method that minimizes the discrepancy between real-case and lab data using joint generalized sliced WD.

Overall, feature-based DTL methods focus on transferring relevant features from a source domain to a target domain, enhancing fault diagnosis in rotating machines. This is achieved by minimizing feature distribution differences across domains through feature transformations. Techniques like TCA, JDA, DAN, and AdaBN are commonly employed for feature extraction and domain adaptation, aligning source and target features to improve diagnostic accuracy in previously unseen, unlabeled conditions. Consequently, feature transfer-based diagnostic models, which facilitate cross-domain feature extraction and fault classification, are extensively researched. This approach reduces reliance on large labeled datasets, making it both efficient and cost-effective. However, current studies mainly address fault diagnosis under varying conditions of the same machine, with limited exploration of knowledge transfer across different machines or domains with significant discrepancies. Therefore, the method's success depends on minimal domain differences and effective feature selection.

4.3 | Parameter Transfer-Based Fault Diagnosis Methods for Rotating Machines

Parameter-based DTL, also known as pretrained model transfer, retains key parameters from a source-trained network while fine-tuning the remaining layers with labeled target data. This approach leverages the idea that early layers capture generalizable features, while later layers focus on task-specific patterns. By adapting pretrained parameters and hyperparameters, deep neural networks (DNNs) can extract similar feature representations across related domains [24, 105, 106]. The pretrained model serves as a foundation for further training on target domain data and enhances learning efficiency. Widely applied in fields like computer vision and natural language processing, this approach

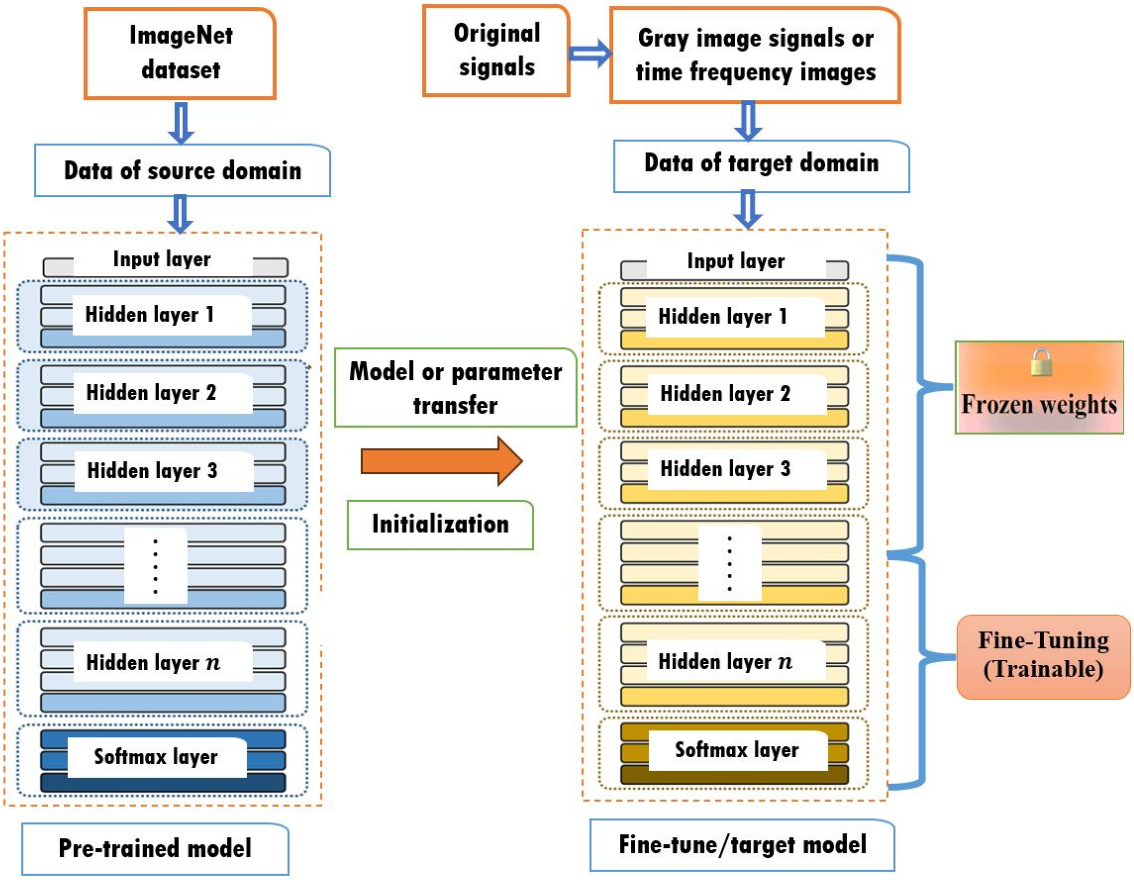

FIGURE 6 | General architectural pipeline for fault diagnosis using fine-tuned pretrained models in DTL methods.

underpins models such as BERT [107] and GPT-3 [108], which, based on Transformer architecture, are fine-tuned for tasks like translation, content generation, and summarization [109]. Similarly, in computer vision, it has significantly improved image processing tasks across diverse domains [110].

Building on these advancements, parameter-based DTL has achieved significant success in diagnosing faults in rotating industrial machines, including bearings [111], gearboxes [112], pumps [113], and turbines [114]. As illustrated in Figure 6, this method applies TL by leveraging pretrained models or transferable parameters for fault diagnosis. The process involves two key steps: first, a diagnostic model is pretrained on a large labeled source domain dataset; then, a small labeled target domain dataset is used to fine-tune the model, refining parameters for task-specific performance. To formalize this approach, the mathematical formulations are defined as follows:

The pretrained model extracts general features from the source dataset D_s. The lower layers capture basic features, while deeper layers learn high-level representations. Hence, the pretrained model is represented as:

$$f(x;\theta) \tag{15}$$

where x denotes the input feature (gray image signals or time–frequency images), and θ represents the parameters learned from the source domain.

Next, fine-tuning adapts the pretrained model to the target domain D_t. Small adjustments suffice for similar datasets, while larger differences require more extensive updates. The fine-tuning objective is given by:

$$\min_{\theta} L(f(x\theta), y) \tag{16}$$

where y represents the true labels in the target domain, and $L(\cdot)$ is the task-specific loss function.

Regularization prevents overfitting, particularly for small target datasets and ensures smooth adaptation. The regularization-enhanced objective is defined as:

$$\min_{\theta} L(f(x\theta), y) + \lambda R(\theta) \tag{17}$$

where λ represents the regularization coefficient, and $R(\theta)$ denotes the regularization function.

Subsequently, freezing lower layers reduces the number of trainable parameters, accelerates training, and preserves generalizable features from the source domain. Only higher layers are updated for task-specific adaptation. The layer-freezing strategy is formulated as:

$$\min_{\theta_t} L\left(f\left(x;\theta_f,\theta_t\right), y\right) \tag{18}$$

where θ_f denotes the parameters of frozen layers, and θ_t represents the parameters of trainable layers.

Most pretrained models used in model-based DTL are DNNs, particularly those derived from deep CNNs, such as AlexNet [115], VGGNet [116], ResNet [117], Xception [118], Inception [119], and EfficientNet [120]. These models are widely adopted due to their hierarchical structure and strong feature extraction capabilities, enabling effective knowledge transfer from image processing to fault diagnosis, even across significantly different domains. Lower convolutional layers capture basic features like edges and curves, while deeper layers learn high-level abstractions essential for diverse applications. Fine-tuning typically involves adjusting weights in the higher layers, with minimal updates needed for similar datasets and broader modifications required for datasets with greater differences. Consequently, pretrained CNNs have demonstrated remarkable success in rotating machine fault diagnosis, particularly when adapted from natural image datasets to time-frequency mechanical data.

Hence, in recent years, researchers have successfully applied model-based transfer techniques, such as AlexNet, VGGNet, ResNet, Xception, Inception, and EfficientNet, for rotating machine fault diagnosis. For example, utilizing AlexNet, Wang et al. [121] investigated the impact of destination domain input images generated using time-frequency methods on a transferred AlexNet model for bearing fault diagnosis, while Cao et al. [106] introduced a CNN-based TL method leveraging AlexNet's pretrained models for gear fault diagnosis. Similarly, Ma et al. [115] employed frequency slice wavelet transform for time-frequency images, which were then used to fine-tune a pretrained AlexNet for bearing diagnosis. Moreover, Jha et al. [122] developed a DTL approach for bearing fault diagnosis using a pretrained AlexNet model, requiring tuning for adaptation across different image classification tasks. Additionally, using VGGNet, Shao et al. [123] utilized pretrained VGG-16 to extract features from time-frequency images in the target domain for machine fault diagnosis. Similarly, Zhou et al. [116] demonstrated improved performance by transferring the VGG-19 pretrained model to diagnose gearbox datasets. In addition, Wen et al. [124] proposed TranVGG-19, a TL approach based on pretrained VGG-19 for bearing fault diagnosis, while Su and Wang [125] developed an optimized VGG-16-based TL approach for rolling bearing fault diagnosis. To improve fault diagnosis accuracy, Ibrahim et al. [126] designed an advanced approach leveraging DTL with VGG-19, integrating it to effectively diagnose single and multiple induction motor faults. Furthermore, Saha et al. [127] developed an advanced fault diagnosis approach for bearings by integrating a VGG-16-based DTL model with a random forest classifier. The model dynamically adapts to diverse working environments and optimizes fault classification across varying operational conditions.

Building on research progress in pretrained models for rotating machine fault diagnosis, the ResNet model has also been effectively applied. For instance, Wen et al. [128] transformed time-domain signals into RGB images to fine-tune a ResNet-50 model for motor bearing and centrifugal pump fault diagnosis. Likewise, Chen et al. [117] proposed a bearing diagnosis method based on time-frequency analysis and TL with ResNet-50, utilizing the CWT. In addition, Kumar et al. [129] proposed a DTL-based method for bearing fault detection in motors, utilizing a pretrained ResNetV2 model, while He et al. [130] developed a multidimensional normalized ResNet for fault diagnosis of rotating machinery. Additionally, Yu et al. [131] proposed a ResNet–Transformer-based domain adaptation model that integrates local and global time–frequency features for rotating machinery fault diagnosis. Moreover, Liu et al. [132] proposed a novel fault diagnosis algorithm for rotating machinery, called the deep adaptation ResNet, which integrates a pretraining model. Zhang et al. [133] proposed a parallel separable ResNet with attention for bearing fault diagnosis, while Jaiswal et al. [134] analyzed deep groove ball bearings using ResNet50 and AlexNet50. Furthermore, Ziad et al. [135] investigated electrical fault detection in permanent magnet synchronous motors by generating current signal data and using it to train a ResNet-50 model.

On the other hand, ongoing research has expanded the use of parameter-based TL in the field. For instance, Zhang et al. [118] introduced a DTL framework integrating CNNs and TL with Xception for machine fault diagnosis. In addition, to maintain deep architecture and efficient memory usage, Jung et al. [136] developed a rotor fault diagnosis algorithm that utilizes CNNs with Xception to detect faults. Additionally, other researchers have utilized Inception to boost detection accuracy. For example, Xu et al. [137] developed a thermography-based motor fault detection method using the InceptionV3 model, enhanced by contrast-limited adaptive histogram equalization and a squeeze-and-excitation channel attention mechanism. Likewise, Liu et al. [119] leveraged a novel fault diagnosis method by implementing model-based TL in the Inception-ResNet-v2 model. In addition, Wei et al. [138] developed a CNN–LSTM diagnostic model enhanced with an improved Inception module, while Li et al. [139] proposed a motor fault diagnosis method integrating an improved Inception module with convolutional block attention module and an improved bi-directional gate recurrent unit. Moreover, the EfficientNet pretrained model has been effectively applied. For instance, Li et al. [140] introduced a planetary gearbox fault diagnosis model that combines gramian angular fields image coding technology with the TL-EfficientNet network model. Similarly, Chen et al. [120] developed an ultra-low-speed bearing recognition model that employs EfficientNet for feature extraction from acoustic emission signals. In addition, Hu et al. [141] proposed a noise-resilient bearing fault diagnosis model based on EfficientNet with an attention mechanism.

Furthermore, to enhance the diagnostic performance of pretrained models in DTL, Zhong et al. [114] developed a TL framework for gas turbine fault diagnosis by transferring a CNN trained on large-scale normal datasets. Likewise, Han et al. [142] proposed a TL framework for diagnosing unseen machine conditions by fine-tuning a CNN trained on large datasets, while Hemmer et al. [105] pretrained a CNN on ImageNet to extract features from wavelet transform images, which were subsequently processed by a sparse autoencoder-based SVM for bearing fault classification. In addition, Shao et al. [123] converted raw signals into images using WT and fine-tuned the upper layers of a CNN pretrained on ImageNet, while Chen et al. [143] introduced a parameter transfer approach using a wide-kernel 1DCNN to extract transferable features for rotating machinery fault diagnosis. Additionally, Hasan et al. [144] employed acoustic spectral images of AE signals to assess mechanical health and proposed a CNN-based parameter transfer method for bearing fault diagnosis. Moreover, Xing et al. [145] proposed a TL-based fault diagnosis method using a pretrained ConvNeXt model, enhancing fault classification under varying conditions with small sample sizes, while Aduwenye et al. [146] developed a robust parameter-based TL method for bearing fault diagnosis, improving adaptability across varying operational and cross-domain conditions with limited data. Furthermore, Zhao et al. [147] introduced an ensemble adaptive batch-normalized CNN with parameter-driven DTL for rotating machinery fault diagnosis, integrating BN and an exponentially decaying learning rate to mitigate internal covariate shift and enhance diagnostic accuracy.

Overall, parameter or pretrained model-based DTL leverages knowledge from a source domain by transferring model parameters, such as weights and feature representations, to a target domain. This approach utilizes pre-trained models, trained on large datasets, where the target model shares the structure, parameters, and hyperparameters of the source model. Commonly applied in DNNs, particularly CNNs, this method transfers lower-level weights from the pre-trained model to the target model, with higher-level weights fine-tuned for the specific fault diagnosis task. This reduces the number of parameters that need updating, accelerating training and improving the target model's performance. In fault diagnosis for rotating machines, pretrained CNNs extract transferable features from sensor data, improving diagnostic accuracy while reducing the need for extensive labeled data. Techniques such as fine-tuning and layer adaptation help adjust pretrained models to new conditions without losing prior knowledge. Compared to training models from scratch, using pretrained weights significantly speeds up the process by minimizing the number of parameters that require training. However, these methods require the target data to be at least partially labeled for fine-tuning. If the source and target data distributions differ significantly, fine-tuning alone cannot yield satisfactory results, requiring feature alignment. While model transfer offers efficiency and reduces training time, its effectiveness relies on the generalizability of the model and the similarity between source and target domains to avoid negative transfer. Consequently, the efficiency of parameter-based DTL models is highly dependent on the availability of labeled samples in the target domain, which poses a challenge in real-world industrial applications due to data scarcity.

4.4 | Adversarial Transfer-Based Fault Diagnosis Methods for Rotating Machines

DTL can be effectively implemented using generative models, leading to the development of adversarial models. When labeled target domain samples are scarce and costly, generative adversarial networks (GANs) offer a solution by generating synthetic labeled data that resemble source domain characteristics. Various GAN-based variants have emerged, significantly enhancing learning performance over traditional DNNs, making GAN-based TL a prominent research area. GANs operate on a game-theoretic framework, consisting of a generator and a discriminator. The generator creates synthetic samples to mimic real data, while the discriminator differentiates between real and generated samples. Their adversarial interaction drives both networks to improve iteratively [148]. As a result, GANs have become widely used across various fields, such as image classification and recognition, to tackle issues related to data limitations and imbalance [149, 150]. Drawing from this success, GANs are increasingly applied to address data imbalance challenges in rotating machine fault diagnosis, significantly improving the diagnostic accuracy for components like bearings [151], gearboxes [152], pumps [153], compressors [154], and motors [155].

Originally, GANs were designed for sample generation and align well with TL, as both aim to ensure similarity between source and target distributions. In adversarial-based TL, instead of generating new samples, the generator extracts feature from source or target domains, treating them as generated representations. The discriminator then distinguishes between these features, encouraging the generator to learn domain-invariant representations. Essentially, the generator functions as a feature extractor, ensuring that target features closely align with the source domain, facilitating efficient TL [156]. As a result, GAN-based DTL, known as data expansion, enhances target domain training by generating additional labeled samples. Adversarial DTL, particularly adversarial domain discriminative adaptation network (ADDA) training, establishes a shared feature space across domains by minimizing task loss in the source domain while maximizing domain confusion loss [33]. As illustrated in Figure 7, ADDA generally consists of three core components: a feature extractor (F), a label classifier (C), and a domain discriminator (D). The discriminator is trained to distinguish between source and target domain features, while the feature extractor simultaneously learns to generate domain-invariant representations that mislead the discriminator. In parallel, the classifier minimizes prediction error on the labeled source samples. Through this joint optimization, adversarial domain adaptation extracts transferable features that are indistinguishable across domains, thereby mitigating the inherent challenges of domain shift.

Formally, in a mathematical representation, the source domain D_s is defined as

$$D_s = \left\{ x_i^s, y_i^s \right\}_{i=1}^{n_s} \tag{19}$$

with n_s labeled samples, while the target domain D_t is represented as

$$D_t = \left\{ x_i^t \right\}_{i=1}^{n_t} \tag{20}$$

with n_t unlabeled samples. The overall learning objective of the ADDA network is expressed as:

$$L_{ADDA} = \frac{1}{n_s} \sum_{x_i \epsilon D_s} L_y\left(C\left(F\left(x_i\right)\right), y_i\right) - \frac{\lambda}{n_s - n_t} \sum_{x_i \epsilon D_s} L_d\left(D\left(F\left(x_i\right)\right), d_i\right) \tag{21}$$

In this formulation, $L_y(\cdot)$ represents the classification loss function applied to the labeled source domain, while $L_d(\cdot)$ denotes the domain discrimination loss function. The variables y_i and d_i correspond to the category label and domain label of each sample, respectively. The parameter λ serves as a trade-off factor that balances classification accuracy on the source domain with domain alignment between the source and target domains, ensuring that the extracted features are both discriminative and transferable.

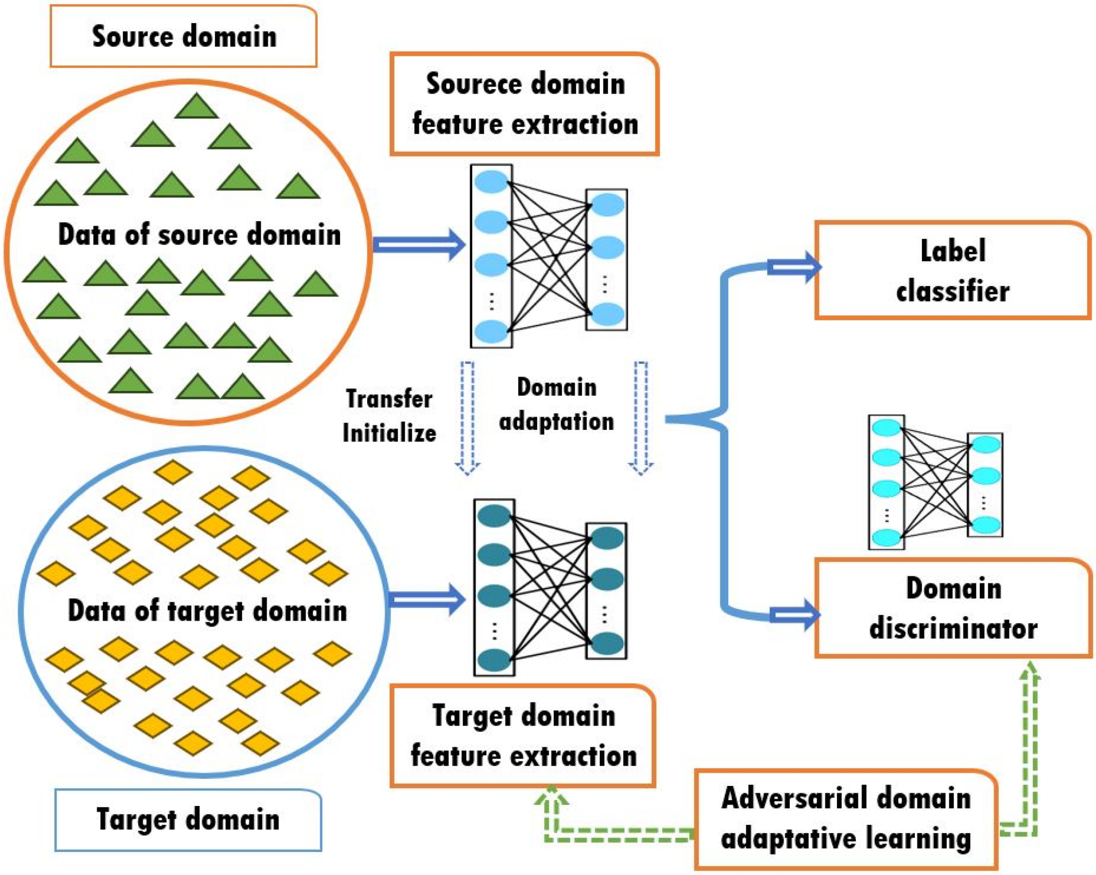

FIGURE 7 | Adversarial domain discriminative adaptation training using transferred feature extractors in source and target models for fault diagnosis.

This adversarial formulation ensures that the classifier learns to minimize task-specific prediction errors, while the feature extractor is trained adversarial to deceive the discriminator. As a result, the network produces robust and domain-invariant feature representations, which significantly improve knowledge transfer and enhance fault diagnosis performance under diverse operating conditions.

In rotating machine fault diagnosis, GAN-based ADDA-adversarial learning focuses on bridging the gap between source and target domains through adversarial training. Many researchers have developed adaptive layers within deep networks to reduce the distribution differences between data features in these domains, enabling the network to adapt effectively and improve diagnostic performance in the target domain. These advancements have led to significant success in the field of domain discriminative-adversarial learning. For example, Li et al. [157] introduced an adversarial adaptive network for bearing diagnosis under variable conditions, while Qin et al. [158] proposed a parameter-sharing adversarial domain adaptation network for planetary gearbox fault diagnosis. Likewise, Jiao et al. [159] used joint maximum mean discrepancy loss in adversarial training to align class identification and domain-invariant features for gear and bearing datasets, and in [160], they introduced a double-level adversarial network that incorporated extractors and classifiers for improved alignment. Moreover, Deng et al. [161] further refined adversarial network performance by adding an attention mechanism for machinery fault diagnosis, while Gao et al. [162] proposed a multi-source domain fusion network leveraging adversarial transfer learning to improve fault diagnosis in rotating machinery. Meanwhile, Gao et al. [163] developed a domain feature decoupling network that separates domain-specific and fault-related features using bearing datasets.

In addition, domain-adversarial neural networks (DANN) have been effectively applied to rotating machinery fault diagnosis as an alternative to ADDA. By incorporating a domain classifier trained adversarially with a gradient reversal layer, DANN enables the extraction of task-discriminative yet domain-invariant features. For example, Xu et al. [164] applied DANN with time–frequency analysis for rotor fault diagnosis, while Wang et al. [165] combined adversarial training, meta-self-training, and an enhanced DANN to achieve robust cross-device fault diagnosis. Likewise, He et al. [166] proposed a DTL-based gearbox fault diagnosis method using discriminative feature extraction and improved DANN, and Chen et al. [167] further advanced this by introducing a triple-improved DANN for bearing fault diagnosis.

Moreover, to improve domain-adversarial learning in rotating machine fault diagnosis, different researchers have applied various improvements and achieved great success. For instance, Xiang et al. [168] introduced a DTL method for bearing fault diagnosis using domain separation reconstruction adversarial networks, which separate domain-difference and domain-invariant features for cross-machine fault diagnosis. Similarly, Shao et al. [169] introduced a deep adversarial TL method for rotating machinery fault diagnosis, utilizing domain adaptation to separately align local and global feature distributions, while Tang et al. [170] presented a fault diagnosis approach for rotating parts in injection molding machines by integrating domain adaptation and adversarial training. In addition, Liu et al. [171] proposed a gas turbine fault diagnosis method using adversarial discriminative domain adaptation TL, transferring pre-trained CNN models from the source to the target domain. Additionally, Chen et al. [172] developed a dual adversarial unsupervised multi-domain adaptation network for fault diagnosis in rotating machinery, while Liu et al. [173] presented a deep multi-source adversarial discrepancy matching adaptation network to improve cross-domain intelligent fault diagnosis accuracy for rotating machinery. Moreover, Wu et al. [174] proposed a conditional distribution-guided adversarial transfer learning network utilizing multi-source domains for bearing fault diagnosis across different machines, while Zou et al. [151] presented a method combining a domain-adversarial network and a signal reconstruction unit to extract domain-invariant features from high-speed train bearing signals. Furthermore, Wang et al. [175] developed a multi-domain collaborative GAN for rotating machinery fault diagnosis, which ensures globally interpretable generation and enhances classification credibility.

Furthermore, building on adversarial learning techniques, research continues to expand in the field. For instance, Guo et al. [176] integrated moment matching with adversarial learning to create a deep convolutional transfer network, and Zhang et al. [177] proposed a multi-adversarial network guided by Wasserstein distance. Likewise, Wang and Liu [178] enhanced the Wasserstein-based adversarial approach with triplet loss for bearing fault diagnosis, and Jiao et al. [179] introduced an unsupervised adversarial adaptation network to achieve cross-domain fault diagnosis without the need for a domain discriminator. Additionally, Liu et al. [180] introduced a semi-supervised joint adaptation TL utilizing conditional adversarial learning for rotary machine fault diagnosis, extracting fault features from unlabeled data by generating pseudo-labels via threshold filtering. Moreover, Li et al. [181] presented an innovative unsupervised TL framework that combines joint distribution and adversarial networks to diagnose faults in bearings and gears of rotating machinery, while Su et al. [182] developed a robust multi-adversarial DTL for multi-source open-set fault diagnosis, effectively handling category shifts in rotating machinery. Furthermore, Wang et al. [183] proposed an enhanced DTL method, leveraging a hybrid domain adversarial learning strategy to improve fault diagnosis in rotating machinery within nuclear power plants.

Overall, adversarial-based DTL has proven effective for fault diagnosis in rotating machines, especially when there are significant domain differences between source and target datasets. While these methods do not require target labels, reliance on data scale and computational resources is high. The GAN-based DTL approach aids in data augmentation by generating auxiliary samples that resemble real ones, enhancing training in the target domain. In particular, ADDA transfer training streamlines domain adaptation by integrating it into network parameter learning and eliminating the need for manual statistical distance measures. This process reduces domain shifts by training a domain discriminator alongside the feature extractor, improving feature representations and ensuring robustness and generalizability across various machine health states and operating conditions. As a result, adversarial DTL addresses label scarcity by enabling knowledge transfer without extensive labeled target data. However, challenges persist: the inclusion of a domain discriminator increases computational costs, and performance depends on the successful convergence of the min-max game between the feature extractor and discriminator, which is difficult to achieve on small datasets. Additionally, methods are sensitive to hyperparameters, and achieving optimal domain alignment requires careful selection of the feature extraction layer. Despite these complexities, adversarial-based methods offer significant potential for improving diagnostic accuracy and addressing complex fault diagnosis tasks, such as label inconsistency. With sufficient training data, high diagnostic performance is consistently achieved across various machines and conditions, holding considerable promise for enhancing adaptability in rotating machine fault diagnosis.

5 | Observation, Research Challenges, and Future Directions

5.1 | Observation

This review discusses a TDL-based approach for fault diagnosis in rotating machines, emphasizing its effectiveness and robustness in knowledge transfer. A comprehensive analysis of existing literature reveals that both traditional ML and DL methods are widely employed in industrial machine fault diagnosis. However, traditional ML techniques face critical limitations. Manual feature extraction limits the handling of complex, high-dimensional data, while shallow models hinder performance in large-scale industrial applications. The separation of feature extraction and decision-making increases computational time and reduces efficiency. Additionally, traditional ML struggles with dynamic, high-dimensional data, leading to suboptimal fault diagnosis as sensor diversity and machine complexity grow. In contrast, DL methods utilize multilayer nonlinear mapping to automatically extract abstract features from diverse mechanical signals, enabling end-to-end fault diagnosis in rotating machines. Their advancement is driven by internal factors such as feature learning, advanced data processing, and architectural innovations, as well as external drivers like the expansion of industrial big data, hardware advancements, and rising industrial demands. Despite progress in DL-based fault diagnosis for rotating machines, two key challenges remain: reliance on labeled data for training and the assumption that training and testing data share the same distribution. In real-world applications, meeting these conditions is difficult. Labeling vast condition monitoring datasets is labor-intensive and costly, while variations in load, speed, and environmental conditions lead to distribution shifts in machine data. These challenges hinder model accuracy, generalizability, and the widespread adoption of DL-based fault diagnosis in rotating industrial machinery.

To overcome these challenges, TL has recently been integrated into rotating machine fault diagnosis, enabling the transfer of knowledge from a source task to a closely related target task.

Specifically, DTL has emerged as a promising new machine learning paradigm that combines the feature representation strengths of DL with the knowledge transfer capabilities of TL. As a result, DTL methods enhance the reliability, robustness, and applicability of DL-based fault diagnosis in rotating machines, driving significant advancements in the field. Moreover, by transferring knowledge from related tasks and adapting to data variations, DTL enables machine fault diagnosis with minimal or no additional labeled data, eliminating the need to retrain models and reducing dependence on extensive labeled datasets.

Consequently, this review paper provides an in-depth analysis of the applications and effectiveness of DTL in intelligent fault diagnosis, emphasizing recent developments in the field. The authors categorized DTL methods for rotating machine fault diagnosis into four key approaches: instance-based, feature/mapping-based, parameter/pretrained model-based, and adversarial-based transfer. The paper begins by introducing the fundamental concepts of DTL, offering clear explanations, and then delves into recent research applications, providing valuable perspectives into various tasks within the domain. Finally, a comprehensive summary is presented to guide the selection of the most appropriate DTL method for rotating machine fault diagnosis applications. Table 1 presents a brief description, characteristics, and a comparison of the strengths and weaknesses of DTL methods in this domain.

From the provided review and Table 1, instance-based transfer methods improve target-domain performance by selectively incorporating relevant source-domain instances into the target dataset. This approach is straightforward to implement and requires minimal labeled data, making it suitable for scenarios with limited fault samples. However, its effectiveness is highly sensitive to the similarity between source and target domains. Moreover, in large-scale industrial systems, managing and processing large volumes of source–target instances can increase computational overhead, particularly when real-time fault diagnosis is required across multiple machines under varying operational and loading conditions.

Feature/mapping-based transfer methods address domain discrepancies by aligning feature representations from the source and target domains within a shared latent space. These methods are highly versatile and maintain stable performance across significant domain shifts, which is particularly advantageous in industrial environments where machines can operate under diverse loads and speed scenarios. However, achieving effective feature alignment often involves substantial computational effort, including identifying optimal transformation functions and calculating distance metrics between features. This huge computational demand can limit scalability when applying these methods to large-scale industrial systems with multiple machines, high-dimensional sensor data, and real-time monitoring requirements.

Parameter/pretrained model-based transfer methods leverage pre-trained models to reduce both training time and computational costs. By utilizing knowledge from source domains, these methods require fewer labeled samples in the target domain and allow flexible fine-tuning to adapt to new tasks. This makes them highly attractive for industrial applications where fault data is limited. However, despite these successes, performance can degrade significantly when substantial disparities exist between the source and target domains, such as differences in machine types, operating conditions, and sensor configurations. In addition, these methods are constrained by input data compatibility, as the architecture and feature requirements of the pre-trained model must align with the target dataset, limiting flexibility in heterogeneous industrial systems. Furthermore, scaling these approaches to large industrial environments can also require substantial memory and processing resources, especially when multiple machines and high-frequency vibration data are involved.

Adversarial-based transfer approaches focus on extracting domain-invariant features through adversarial training, making them suitable for semi-supervised or unsupervised learning scenarios. These methods can enable knowledge transfer even with limited labeled data, addressing common challenges in industrial fault diagnosis, such as imbalanced health states and varying operational scenarios. However, training adversarial models is computationally intensive and requires careful tuning to achieve stability and convergence. Real-time deployment in complex industrial settings remains challenging due to high computational demands, the need for rapid inference, and the difficulty of handling heterogeneous data from different machines or sensors.

Overall, while rotating machine fault diagnosis based on these four categorized DTL methods offers strong potential for industrial applications, careful consideration of computational requirements, domain alignment, efficient instance selection, and scalable feature alignment strategies is essential for successful large-scale implementation in diverse and dynamic industrial environments.

Moreover, to strengthen the impact of this review and provide practitioners with clear guidance in selecting DTL methods for specific fault diagnosis scenarios, Table 2 presents a detailed comparison of each DTL variant within the four categorized methods. This comparison emphasizes the strengths and limitations of each approach, providing practical considerations and a concise description of each method to support informed decision-making. Grounded in the categorizations presented in the review, findings from existing literature, and perspectives drawn from the authors' domain expertise, the table serves as a comprehensive guide for researchers, industrial experts, policymakers, and engineers. This table provides nuanced guidance on selecting the most appropriate DTL approach in the fields. As a result, the table facilitates well-informed decisions for implementing intelligent fault diagnosis in rotating machines under diverse operational and loading conditions.

Specifically, instance-based methods such as TrAdaBoost and importance sampling/weighting focus on selectively reusing relevant source domain instances to enhance target performance. In addition, feature-based methods, including TCA, JDA, and DAN, address domain discrepancies by aligning features in a shared latent space. Moreover, parameter-based methods leverage DNNs such as AlexNet, VGGNet, ResNet, Xception, Inception, and EfficientNet to efficiently transfer learned parameters to new target tasks. On the other hand, adversarial transfer methods, such as DANN and ADDA, employ adversarial training to

TABLE 1 | The description, characteristics, strengths, and weaknesses of each type of TDL method for rotating machine fault diagnosis.

DTL method	Short description and characteristics of each method	Strengths	Weaknesses
Instance-based transfer	Description: Enhancing target data by incorporating instances from the source domain into the target domain. Characteristics: Improves target domain performance by selecting and reusing relevant labeled data from a source domain with similar distributions, effectively augmenting the target dataset.	1. The approach is straightforward and user-friendly. 2. The integrated model demonstrates robustness. 3. Enables knowledge transfer with minimal labeled data.	1. Performance drops with mismatched domains. 2. Requires careful instance selection to avoid noise. 3. Limited transfer samples are available in practice. 4. Minimal disparities between domains are necessary.
Feature/mapping-based transfer	Description: Mitigates feature discrepancies between source and target domains by aligning their representations in a shared latent space. Characteristics: Identifies transformation functions to map features from both domains into a common feature space, facilitating effective knowledge transfer.	1. Remarkable capacity for feature extraction. 2. Maintains robust performance even when significant differences exist between domains. 3. Suitable for complex data distributions with high versatility in application.	1. There is a need for consistency in the feature space between the source and target domains. 2. Involves substantial computational effort and poses challenges in determining appropriate feature mappings between domains. 3. Challenges in selecting an appropriate distance metric.
Parameter/pretrained model-based transfer	Description: Adapts a pre-trained model's parameters from a source domain to a target domain. Characteristics: Involves fine-tuning the pre-trained model on target domain data to adjust its parameters for the new task.	1. Minimal need for labeled data. 2. Simplifies the training process, reduces computational time, and is straightforward to implement. 3. High flexibility for fine-tuning. 4. Availability of many free/open-source models.	1. Performance declines as the disparity between source and target domains increases; requires a modest amount of labeled data in the target domain. 2. Fine-tuning and test data must share the same distribution. 3. Constraints on input data format.
Adversarial-based transfer	Description: Utilizes adversarial training to extract indistinguishable feature representations between source and target domains. Characteristics: Addresses complex operational scenarios and imbalanced health states by optimizing domain distributions through adversarial mechanisms.	1. Extracts domain-invariant features, improving generation in the source domain. 2. Facilitates semi-supervised or unsupervised learning, enabling knowledge transfer without extensive labeled data. 3. Eliminates the need for manually designed distance metrics.	1. Challenging to train an adversarial model. 2. Challenges persist in model adaptability; practical scenarios require real-time fault diagnosis under intricate working conditions. 3. Incapable of processing heterogeneous data.

TABLE 2 | Strengths, limitations, and concise descriptions of DTL variants for rotating machine fault diagnosis.

DTL method	Variant	Concise description	Strengths	Weaknesses	References
Instance-based transfer	Transfer AdaBoost/ TrAdaBoost	A boosting-based method that selectively re-weights source domain instances to emphasize those most relevant and improves target task performance.	1. Effective with scarce labeled target data and easy to implement. 2. Minimizes negative transfer by down-weighting irrelevant source instances.	1. Performance drops significantly if target and source targets differ; sensitive to domain mismatch. 2. High computational cost for large-scale machine datasets.	[50–52]
	Importance sampling/weighting	Probabilistic re-weighting of source domain samples based on their similarity to the target distribution.	1. Improves effectiveness and generalization in small target datasets. 2. Can handle partial domain overlap.	1. Requires accurate estimation of source-to-target instance similarity. 2. Performance is limited when distribution divergence is high.	[55, 56]
Feature/mapping-based transfer	TCA	Maps source and target features into a shared subspace (or Hilbert space) to align their marginal distributions and minimize domain discrepancy.	1. Effectively handles moderate domain shifts. 2. Easy to implement for open-access datasets.	1. Requires careful tuning of network parameters, training hyperparameters, and kernel function selection. 2. Limited scalability for high-dimensional data.	[65, 83, 84]
	JDA	Aligns both marginal and conditional distributions between source and target domains in a shared subspace.	1. More effective in handling conditional distribution differences across domains. 2. Minimizes negative transfer compared to TCA.	1. Computationally intensive and sensitive to kernel function selection. 2. Performance depends heavily on the number of target samples.	[89–91]
	DAN	Incorporates MMD or MK-MMD into DNNs to refine feature alignment during training and minimize domain discrepancy.	1. Learns domain-invariant features by combining deep feature extraction with domain adaptation. 2. Effective and generalizable for large-scale and complex fault data.	1. Requires careful tuning of network parameters and training hyperparameters. 2. Risk of overfitting with very small target-domain datasets.	[94, 95]

(Continues)

TABLE 2 | Strengths, limitations, and concise descriptions of DTL variants for rotating machine fault diagnosis.

DTL method	Variant	Concise description	Strengths	Weaknesses	References
Parameter/pretrained model-based transfer	AlexNet	Transfers pretrained parameters for fine-tuning on new vibration or time-frequency fault data; this pioneering CNN architecture advanced DL for fault diagnosis.	1. Reduces training time. 2. Requires fewer labeled target samples while delivering proven performance in fault diagnosis tasks.	1. Sensitive to source-target domain gap. 2. Limited flexibility for heterogeneous inputs and requires large memory.	[115, 122]
	VGGNet	Deep CNN pretrained model with 16–19 layers, using small (3×3) convolutional filters for feature extraction and transfer to target datasets.	1. Wide community support and resources. 2. Easy to implement and fine-tune for new fault diagnosis tasks.	1. Risk of overfitting with small machine fault datasets. 2. Sensitive to input data and requires high computational resources.	[125–127]
	ResNet	Deep residual network enabling very deep architectures (up to 152 layers) through residual connections and allows transfer of learned parameters to new fault diagnosis tasks.	1. Mitigates degradation and gradient vanishing/explosion issues in deep CNNs. 2. Supports very deep architectures and achieves higher accuracy on complex fault datasets compared to AlexNet and VGGNet.	1. Requires careful tuning of network depth and learning rates. 2. High GPU memory demand.	[129–131]
	Inception	Also known as GoogLeNet, employs inception modules (multi-branch CNN architecture) with parallel filters of different sizes to capture multi-scale features for parameter transfer.	1. Efficient computation using parallel filters while capturing multi-scale features. 2. Generalizes well for complex machinery and industrial datasets.	1. Complex architecture makes fine-tuning more challenging. 2. Sensitive to preprocessing of input data.	[119, 138]
	Xception	Utilizes depthwise separable convolutions, splitting standard convolutions into depthwise and pointwise operations, to enable efficient DTL with pretrained CNNs for fault diagnosis.	1. Lightweight and computationally efficient compared to Inception. 2. Flexible for DTL-based fault diagnosis tasks.	1. Requires careful learning rate tuning and is prone to overfitting with small datasets. 2. Limited robustness when the target domain deviates significantly.	[118, 136]
	EfficientNet	Developed by Google, employs a compound scaling method that simultaneously balances network depth, width, and input resolution, using fixed coefficients, unlike earlier pretrained CNNs that scaled each dimension independently.	1. Provides balanced improvements in both accuracy and computational efficiency compared to earlier models. 2. Performs effectively on both large and small datasets.	1. Requires careful tuning of network parameters and hyperparameters. 2. Complex architecture increases implementation difficulty and limits feature interpretability.	[120, 140]

(Continues)

TABLE 2 | (Continued)

DTL method	Variant	Concise description	Strengths	Weaknesses	References
Adversarial-based transfer	ADDA	Adversarially trains a feature extractor against a domain discriminator, while a classifier minimizes source prediction error to achieve cross-domain transfer.	1. Flexible architecture for different domains and reduces negative transfer. 2. Effective and robust under heterogeneous source–target domain conditions.	1. Training can be unstable due to adversarial optimization and is computationally expensive. 2. Requires careful tuning of the discriminator and generator.	[157–159]
	DANN	Introduces a domain classifier with a gradient reversal layer to extract features that are both task-discriminative and domain-invariant in a single end-to-end framework.	1. Robust extraction of domain-invariant features. 2. Supports semi-supervised learning and performs well with limited labeled target data.	1. Training instability due to adversarial optimization. 2. High computational cost and sensitivity to hyperparameter tuning.	[164–166]

extract domain-invariant features, enabling effective knowledge transfer under complex and partially labeled scenarios.

5.2 | Research Challenges and Future Directions

Despite significant progress in DTL and its various methods for diagnosing faults in rotating machines, certain challenges persist, requiring further investigation. Based on the categorizations presented in this review, perspectives drawn from prior studies, and the authors' expertise in the field, key challenges are identified. In addition, the corresponding future research directions are outlined to support researchers, engineers, and practitioners as follows:

1. Most existing DTL-based fault diagnosis methods rely on single-channel vibration signals, which not effectively capture non-structural faults. This limitation underscores the need for multi-sensor data fusion, integrating diverse data types such as acoustics signals, temperature, current, and thermal imaging to provide a more comprehensive assessment of machinery health. The diagnostic effectiveness of each signal varies under different operating conditions, limiting the performance of single-modal models. Additionally, single-source datasets are susceptible to external noise and interference, hindering the extraction of complete fault features. Despite these challenges, research on applying DTL to multi-source heterogeneous data in industrial fault diagnosis remains limited.

Advancing multi-source DTL strategies is essential for improving fault diagnosis. This involves leveraging multiple sensor locations and diverse sensor types to develop robust fusion-based diagnostic models that offer a holistic view of equipment health. Future research should prioritize optimizing cross-domain learning techniques and integrating unsupervised and weakly supervised domain transfer learning to enhance diagnostic accuracy and adaptability in complex industrial environments.

2. Addressing the negative transfer phenomenon remains a significant challenge in DTL-based diagnostic models, particularly in scenarios with high domain discrepancies where the risk of ineffective knowledge transfer increases. Assessing the transferability of knowledge or data is essential for developing accurate and robust DTL models, as it helps prevent negative transfer and ensures the selection of relevant diagnostic knowledge from the source domain.

Consequently, mitigating negative transfer is a key research priority, requiring the development of advanced strategies to enhance the effectiveness of DTL in scenarios with substantial domain differences. In addition, future research should focus on establishing systematic criteria for evaluating transferability in DTL, which could improve its applicability across diverse industrial settings. Developing a standardized distance measurement criterion or a general diagnostic model could improve its applicability across diverse industrial settings.

3. Most current DTL-based fault diagnosis methods for rotating machinery rely on knowledge transfer from a

single source domain, following a one-source-to-one-target paradigm. However, this approach is restrictive in industrial settings, where a single source domain can provide insufficient data for effective learning. In addition, challenges such as data privacy and security further complicate knowledge transfer across different domains.

Federated transfer learning (FTL) presents a promising solution by leveraging multiple source domains to enhance learning while preserving data privacy. In addition, developing fault diagnosis methods based on FTL can improve model generalization and robustness in industrial applications. Future research should focus on optimizing FTL frameworks for secure and efficient multi-source knowledge transfer in rotating machinery fault diagnosis.

4. DTL methods focus on selecting and transferring only relevant knowledge from the base model to new tasks, utilizing techniques such as selective fine-tuning and distillation to extract critical features from source domains. However, the design and selection of network parameters, including architecture and fine-tuning hyperparameters, remain largely subjective, with no standardized rules for determining the most suitable parameters. This lack of standardization leads to reliance on empirical methods or trial-and-error approaches. Hence, the intricate structure of DTL models demands intelligent optimization of both network and training parameters, a process currently unsupported by a comprehensive theoretical framework.

Future research should focus on developing dynamic optimization strategies to tackle these challenges. Techniques such as particle swarm optimization and genetic algorithms can aid in identifying optimal parameter configurations. In addition, integrating automated machine learning could streamline hyperparameter selection, while implementing self-tuning mechanisms for neural networks can further enhance training efficiency. Moreover, investigating the link between parameter selection and mechanical signal characteristics could offer valuable perspectives for improving fault diagnosis models.

5. In existing DTL-based rotating machine fault diagnosis, most publicly available fault diagnosis datasets are collected under controlled laboratory conditions, where fault characteristics are distinct and easily identifiable. However, real-world industrial environments involve varying loads, noise, and external disturbances, making fault diagnosis significantly more complex. Laboratory simulations fail to fully replicate these challenging conditions, leading to discrepancies in data distribution between experimental and practical settings. In addition, existing DTL approaches primarily focus on knowledge transfer between closely related domains, such as different operating states of the same machine. Generalizing across vastly different domains, such as transferring knowledge between bearings and gearboxes, remains an unresolved challenge.

Advancing cross-domain knowledge transfer in DTL requires new strategies to bridge the gap between laboratory simulations and real-world industrial applications. Future research should prioritize developing robust signal processing techniques capable of extracting meaningful fault features under high-noise conditions. In addition, exploring methods to facilitate effective knowledge transfer across dissimilar machine components will be crucial for enhancing the adaptability and accuracy of DTL-based machine fault diagnosis models in practical scenarios.

6. Current DTL methods are predominantly designed for fault diagnosis in individual rotating machines. However, with the rise of clustered machinery systems and the continuous development of new equipment in modern manufacturing, the limitations of single-machine-focused models have become more apparent. Ensuring that DTL models can generalize effectively to both newly developed and similar machines remains a critical challenge.

Future research should focus on enhancing the generalization capabilities of DTL models to enable seamless adaptation to different yet related machines. Developing robust DTL frameworks that efficiently handle variations across multiple machines will be essential for advancing fault diagnosis in large-scale industrial settings.

7. Existing methods typically assume that data labels are entirely accurate, but in practical applications, machine data often suffer from labeling errors due to factors such as human mistakes, environmental noise, and instrumentation inaccuracies. Manually verifying and recalibrating these labels can be costly or even infeasible. Incorrect labels lead to erroneous gradient information, which undermines the performance of diagnostic and predictive models. Furthermore, label noise can result in negative transfer, complicating fault diagnosis.

Addressing the issue of noisy labels is essential for improving the effectiveness and robustness of DTL models. Future research should focus on developing strategies to mitigate label noise, ensuring more reliable transfer learning for rotating machine fault diagnosis. This will enhance diagnostic accuracy and prevent the negative transfer effects caused by mislabeled data.

8. Current DTL models are typically tested in offline settings, which fail to account for the real-time fluctuations in equipment status observed in industrial applications. Real-world scenarios demand models capable of adapting online, as continuous data streams reflect ongoing changes in machine health. Moreover, the aging of equipment or shifts in operating conditions can lead to distribution shifts in the data, ultimately reducing model accuracy.

To address these challenges, future research should prioritize the development of online transfer learning strategies that facilitate real-time feature selection and self-adaptation within DTL frameworks. In addition, implementing an online re-learning mechanism to fine-tune model parameters using freshly collected data will enable continuous updates, ensuring sustained accuracy even as equipment ages or operational conditions evolve. This will allow for more timely fault detection and robust performance over time.

9. Bridging the gap between research and practice in the fault diagnosis of rotating machines using DTL methods remains a critical challenge. Key obstacles include limited access to

industrial data due to proprietary restrictions, the scarcity of fault data compared to abundant health data, variability in operating and loading conditions across different machines, and the lack of standardized protocols for validating and benchmarking fault diagnosis models. These challenges hinder the practical implementation and generalization of advanced methods, including DTL, in real-world industrial settings.

To address these issues, future research should focus on fostering collaborations between academia and industry to enable large-scale experimental validations, developing methods for handling imbalanced datasets and health classes, and performing systematic benchmarking across diverse operational scenarios to create effective and robust DTL frameworks. In addition, combining theoretical advances with practical deployment considerations, such as real-time monitoring, adaptability to changing conditions, and interpretability of model outputs, will be essential to translate research innovations into effective, reliable, and industry-ready fault diagnosis solutions.

10. Ensuring the stability of DTL models remains a critical challenge in rotating machinery fault diagnosis. Stability issues arise from the non-stationary, nonlinear, and noisy nature of vibration and other health-monitoring data in rotating machines, which can lead to overfitting, gradient vanishing/exploding, and inconsistent predictions under varying operating and load conditions. In DTL scenarios, domain mismatches between source and target data further aggravate instability, while sensitivity to minor perturbations and limited robustness to operational variations reduce the reliability of diagnostic outputs.

To address these concerns, future research should focus on integrating theoretical stability frameworks, such as Lyapunov-based criteria, into model design, particularly for recurrent and higher-order neural networks. Recent advanced architectures, including Memory Recurrent Elman Neural Networks [12], *Lyapunov-stability-based Context-Layered Recurrent Pi-Sigma Networks* [13], *and Double Internal Loop Higher-Order Recurrent Neural Networks* [14], *demonstrate potential in guaranteeing stability while preserving learning efficiency. In addition, stability-aware training strategies, such as regularization techniques, adaptive learning rate schemes, and gradient clipping, can improve model resilience against perturbations and domain shifts. Therefore, investigating the link between model parameters, signal characteristics, and stability, along with incorporating online adaptation mechanisms, will be essential to develop robust, reliable, and generalizable DTL frameworks for real-world industrial fault diagnosis.*

11. Relying solely on DTL is insufficient for developing robust AI solutions for fault diagnosis in rotating industrial machines. To address the complexities of real-world industrial environments, future strategies should integrate other machine learning approaches, such as continuous learning, meta-learning, federated learning, incremental learning, and evolutionary learning. These technologies can equip models with self-learning abilities, enhancing the adaptability of DTL models for more efficient and cost-effective industrial applications. Additionally, while large models like transformers and diffusion models have shown promise in fault diagnosis, their effectiveness depends on the nature of the machine data and the specific requirements of the fault diagnosis task.

To improve fault diagnosis models, future research should focus on integrating emerging ML techniques with DTL, enhancing the self-adaptive capabilities of models. Furthermore, the computational efficiency and interpretability of large models should be carefully considered, especially in real-time or resource-constrained industrial environments. The application of these advanced techniques, combined with a careful assessment of data characteristics and task requirements, can significantly boost the performance and applicability of fault diagnosis solutions in the field.

6 | Conclusions and Perspectives

This review comprehensively explores the development of DTL for enhancing intelligent fault diagnosis in rotating machines across various industrial sectors. The paper begins by systematically defining DTL and providing the necessary theoretical foundation, setting the stage for a deeper understanding of DTL methods. It then delves into recent research applications, offering valuable perspectives into various tasks within the domain. In this review, the authors categorize the existing DTL methods in rotating machine fault diagnosis into four primary groups: instance-based, feature-based, parameter-based, and adversarial-based transfer. Each category is discussed in detail, highlighting key concepts, representative algorithms, and models. Most importantly, the paper presents the application of these methods to fault diagnosis in rotating machines, drawing from recent studies where DTL has become a research hotspot. It also emphasizes the development of diverse transfer architectures tailored to specific tasks. This review provides a thorough roadmap for both practitioners and researchers, offering detailed guidance for those looking to embark on or expand their work in this field.

In addition, the review offers guidance on selecting the most suitable DTL method for rotating machine fault diagnosis applications, providing concise descriptions, characteristics, and an analysis of the strengths and weaknesses of each approach. By comparing the advantages and limitations of different methods, the paper identifies current challenges faced by DTL in this domain and explores potential future directions for DTL-based fault diagnosis in rotating machines. Addressing these challenges will drive progress in the field, leading to enhanced fault diagnosis accuracy and reliability. This, in turn, will reduce costs, improve operational efficiency, and increase safety in industries such as aviation, smart manufacturing, energy generation, and automotive.

Overall, this review paper serves as a crucial reference for researchers, industry experts, practitioners, and policymakers by offering in-depth perspectives into the latest advancements in DTL for rotating machine fault diagnosis. As discussed, the emerging challenges identified, along with the proposed research directions, provide a foundation for developing innovative strategies to improve fault diagnosis accuracy and reliability in rotating machines. This, in turn, can drive improved performance and reduce operational costs. Moreover, the review emphasizes the importance of considering feasibility, reliability, explainability, and real-world applications when designing DTL systems in this domain.

In future research, the authors plan to propose a novel DTL method to enhance fault diagnosis for rotating machines. This method will focus on transferability assessment to mitigate negative transfer, facilitate transfer across different signal types, and enable multi-source transfer. In addition, it will address challenges associated with transfer under abnormal data quality, improve online TL by leveraging emerging technologies such as automated machine learning and digital twins, and explore novel applications for transfer between different machines. Furthermore, as part of future research, the authors will consider noise and disturbances in fault diagnosis tasks to enhance robustness under realistic noisy and disturbed industrial conditions. By addressing these challenges, the proposed method aims to significantly improve both the accuracy and reliability of fault diagnosis in practical industrial environments.

Author Contributions

Temesgen Tadesse Feisa: writing – review and editing, writing – original draft, visualization, software, methodology, conceptualization, investigation. **Hailu Shimels Gebremedhen:** supervision, resources, investigation, methodology, writing – review and editing. **Fasikaw Kibrete:** investigation, validation, visualization, writing – original draft. **Dereje Engida Woldemichael:** visualization, supervision, investigation, writing – review and editing. **Solomon Dargie Ageze:** investigation, visualization, validation, writing – original draft. **Henok Dereje Nega:** validation, visualization, writing – review and editing, writing – original draft.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. T. Han, L. Zhang, Z. Yin, and A. C. C. Tan, "Rolling Bearing Fault Diagnosis With Combined Convolutional Neural Networks and Support Vector Machine," *Measurement* 177 (2021): 109022, https://doi.org/10.1016/j.measurement.2021.109022.

2. Q. Wang, R. Huang, J. Xiong, et al., "A Survey on Fault Diagnosis of Rotating Machinery Based on Machine Learning," *Measurement Science and Technology* 35, no. 10 (2024): 102001, https://doi.org/10.1088/1361-6501/ad6203.

3. Y. Ding, M. Jia, Q. Miao, and Y. Cao, "A Novel Time–Frequency Transformer Based on Self–Attention Mechanism and Its Application in Fault Diagnosis of Rolling Bearings," *Mechanical Systems and Signal Processing* 168 (2022): 108616, https://doi.org/10.1016/j.ymssp.2021.108616.

4. F. Kibrete, D. E. Woldemichael, and H. S. Gebremedhen, "Optimization of Sample Size, Data Points, and Data Augmentation Stride in Vibration Signal Analysis for Deep Learning-Based Fault Diagnosis of Rotating Machines," *Shock and Vibration* 2025, no. 1 (2025): 5590157, https://doi.org/10.1155/VIB/5590157.

5. A. K. S. Jardine, D. Lin, and D. Banjevic, "A Review on Machinery Diagnostics and Prognostics Implementing Condition-Based Maintenance," *Mechanical Systems and Signal Processing* 20, no. 7 (2006): 1483–1510, https://doi.org/10.1016/j.ymssp.2005.09.012.

6. Y. Guo, Z. Cheng, J. Zhang, B. Sun, and Y. K. Wang, "A Review on Adversarial–Based Deep Transfer Learning Mechanical Fault Diagnosis," *Journal of Big Data* 11, no. 1 (2024): 1–26, https://doi.org/10.1186/s40537-024-01006-4.

7. J. Jiao, M. Zhao, J. Lin, and K. Liang, "A Comprehensive Review on Convolutional Neural Network in Machine Fault Diagnosis," *Neurocomputing* 417 (2020): 36–63, https://doi.org/10.1016/j.neucom.2020.07.088.

8. Y. Liu, H. Gao, L. Guo, A. Qin, C. Cai, and Z. You, "A Data-Flow Oriented Deep Ensemble Learning Method for Real-Time Surface Defect Inspection," *IEEE Transactions on Instrumentation and Measurement* 69, no. 7 (2020): 4681–4691, https://doi.org/10.1109/TIM.2019.2957849.

9. F. Kibrete, D. E. Woldemichael, H. S. Gebremedhen, and T. T. Feisa, *Parallel Hidden Layers Bidirectional Long Short-Term Memory Framework With Hierarchical Attention Mechanisms for Fault Diagnosis of Rotating Machines* (Springer, 2025), 103–122, https://doi.org/10.1007/978-3-031-87352-2_5.

10. S. Almabdy and L. Elrefaei, "Deep Convolutional Neural Network-Based Approaches for Face Recognition," *Applied Sciences* 9, no. 20 (2019): 4397, https://doi.org/10.3390/app9204397.

11. S. M. Anwar, M. Majid, A. Qayyum, M. Awais, M. Alnowami, and M. K. Khan, "Medical Image Analysis Using Convolutional Neural Networks: A Review," *Journal of Medical Systems* 42, no. 11 (2018): 1–13, https://doi.org/10.1007/s10916-018-1088-1.

12. R. Kumar, "Memory Recurrent Elman Neural Network-Based Identification of Time-Delayed Nonlinear Dynamical System," *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 53, no. 2 (2023): 753–762, https://doi.org/10.1109/TSMC.2022.3186610.

13. R. Kumar, "A Lyapunov-Stability-Based Context-Layered Recurrent Pi-Sigma Neural Network for the Identification of Nonlinear Systems," *Applied Soft Computing* 122 (2022): 108836, https://doi.org/10.1016/j.asoc.2022.108836.

14. R. Kumar, "Double Internal Loop Higher-Order Recurrent Neural Network-Based Adaptive Control of the Nonlinear Dynamical System," *Soft Computing* 27, no. 22 (2023): 17313–17331, https://doi.org/10.1007/s00500-023-08061-8.

15. B. Yang, S. Xu, Y. Lei, C. G. Lee, E. Stewart, and C. Roberts, "Multi-Source Transfer Learning Network to Complement Knowledge for Intelligent Diagnosis of Machines With Unseen Faults," *Mechanical Systems and Signal Processing* 162 (2022): 108095, https://doi.org/10.1016/j.ymssp.2021.108095.

16. K. Zhang, Y. Li, P. Scarf, and A. Ball, "Feature Selection for High-Dimensional Machinery Fault Diagnosis Data Using Multiple Models and Radial Basis Function Networks," *Neurocomputing* 74, no. 17 (2011): 2941–2952, https://doi.org/10.1016/j.neucom.2011.03.043.

17. B. G. Mejbel, S. A. Sarow, M. T. Al-Sharify, L. A. Al-Haddad, A. A. F. Ogaili, and Z. T. Al-Sharify, "A Data Fusion Analysis and Random Forest Learning for Enhanced Control and Failure Diagnosis in Rotating Machinery," *Journal of Failure Analysis and Prevention* 24, no. 6 (2024): 2979–2989, https://doi.org/10.1007/s11668-024-02075-6.

18. A. H. M. Ripon and M. A. Ullah, "Rotating Machine Fault Detection Using Support Vector Machine (SVM) Classifier," in *2023 4th International Conference on Computing and Communication Systems, I3CS 2023* (Institute of Electrical and Electronics Engineers Inc, 2023), https://doi.org/10.1109/I3CS58314.2023.10127320.

19. Z. Ren, Y. Tang, and W. Zhang, "Quality-Related Fault Diagnosis Based on k-Nearest Neighbor Rule for Non-Linear Industrial Processes," *International Journal of Distributed Sensor Networks* 17, no. 11 (2021): 155014772110559, https://doi.org/10.1177/15501477211055931.

20. X. Zhang, D. Jiang, Q. Long, and T. Han, "Rotating Machinery Fault Diagnosis for Imbalanced Data Based on Decision Tree and Fast Clustering Algorithm," *Journal of Vibroengineering* 19, no. 6 (2017): 4247–4259, https://doi.org/10.21595/jve.2017.18373.

21. W. K. Ngui, M. S. Leong, M. I. Shapiai, and M. H. Lim, "Blade Fault Diagnosis Using Artificial Neural Network," *International Journal of Applied Engineering Research* 12, no. 4 (2017): 519–526.

22. X. J. Yi, Y. F. Chen, and P. Hou, "Fault Diagnosis of Rolling Element Bearing Using Naïve Bayes Classifier," *Vibroengineering Procedia* 14 (2017): 64–69, https://doi.org/10.21595/vp.2017.19153.

23. T. T. Feisa, H. S. Gebremedhen, F. Kibrete, and D. E. Woldemichael, "One-Dimensional Deep Convolutional Neural Network-Based Intelligent Fault Diagnosis Method for Bearings Under Unbalanced Health and High-Class Health States," *Structural Control and Health Monitoring* 2025, no. 1 (2025): 6498371, https://doi.org/10.1155/stc/6498371.

24. J. Cen, Z. Yang, X. Liu, J. Xiong, and H. Chen, "A Review of Data-Driven Machinery Fault Diagnosis Using Machine Learning Algorithms," *Journal of Vibration Engineering & Technologies* 10, no. 7 (2022): 2481–2507, https://doi.org/10.1007/s42417-022-00498-9.

25. S. Oh, S. Han, and J. Jeong, "Multi-Scale Convolutional Recurrent Neural Network for Bearing Fault Detection in Noisy Manufacturing Environments," *Applied Sciences* 11, no. 9 (2021): 3963, https://doi.org/10.3390/app11093963.

26. F. Kibrete and D. E. Woldemichael, "Applications of Artificial Intelligence for Fault Diagnosis of Rotating Machines: A Review," in *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST* (Springer, 2023), 41–62, https://doi.org/10.1007/978-3-031-28725-1_4.

27. S. Yang, X. Kong, Q. Wang, Z. Li, H. Cheng, and K. Xu, "Deep Multiple Auto-Encoder With Attention Mechanism Network: A Dynamic Domain Adaptation Method for Rotary Machine Fault Diagnosis Under Different Working Conditions," *Knowledge-Based Systems* 249 (2022): 108639, https://doi.org/10.1016/j.knosys.2022.108639.

28. F. Kibrete, D. E. Woldemichael, and H. S. Gebremedhen, *Rotating Machinery Fault Diagnosis Using One-Dimensional Convolutional Neural Network* (Springer Nature Switzerland, 2025), 83–94, https://doi.org/10.1007/978-3-031-77339-6_5.

29. H. Jiang, H. Shao, X. Chen, and J. Huang, "A Feature Fusion Deep Belief Network Method for Intelligent Fault Diagnosis of Rotating Machinery," *Journal of Intelligent Fuzzy Systems* 34 (2018): 3513–3521, https://doi.org/10.3233/JIFS-169530.

30. Y. Hou, J. Ma, J. Wang, T. Li, and Z. Chen, "Enhanced Generative Adversarial Networks for Bearing Imbalanced Fault Diagnosis of Rotating Machinery," *Applied Intelligence* 53, no. 21 (2023): 25201–25215, https://doi.org/10.1007/s10489-023-04870-4.

31. X. Song, D. Zhu, and S. Sun, "A New Fault Diagnosis Model of Rolling Element Bearing Based on a Recurrent Neural Network," *Proceedings of the Institution of Mechanical Engineers, Part E, Journal of Process Mechanical Engineering* 236, no. 4 (2022): 1430–1439, https://doi.org/10.1177/09544089211064464.

32. W. Cui, G. Meng, A. Wang, X. Zhang, and J. Ding, "Application of Rotating Machinery Fault Diagnosis Based on Deep Learning," *Shock and Vibration* 2021, no. 1 (2021): 3083190, https://doi.org/10.1155/2021/3083190.

33. P. Yan, A. Abdulkadir, P. P. Luley, et al., "A Comprehensive Survey of Deep Transfer Learning for Anomaly Detection in Industrial Time Series: Methods, Applications, and Directions," *IEEE Access* 12 (2024): 3768–3789, https://doi.org/10.1109/ACCESS.2023.3349132.

34. C. Li, S. Zhang, Y. Qin, and E. Estupinan, "A Systematic Review of Deep Transfer Learning for Machinery Fault Diagnosis," *Neurocomputing* 407 (2020): 121–135, https://doi.org/10.1016/j.neucom.2020.04.045.

35. M. Hakim, A. A. B. Omran, A. N. Ahmed, M. Al-Waily, and A. Abdellatif, "A Systematic Review of Rolling Bearing Fault Diagnoses Based on Deep Learning and Transfer Learning: Taxonomy, Overview, Application, Open Challenges, Weaknesses and Recommendations," *Ain Shams Engineering Journal* 14, no. 4 (2023): 101945, https://doi.org/10.1016/j.asej.2022.101945.

36. J. Yu and Y. Zhang, "Challenges and Opportunities of Deep Learning-Based Process Fault Detection and Diagnosis: A Review," *Neural Computing and Applications* 35, no. 1 (2023): 211–252, https://doi.org/10.1007/s00521-022-08017-3.

37. H. Chen, H. Luo, B. Huang, B. Jiang, and O. Kaynak, "Transfer Learning-Motivated Intelligent Fault Diagnosis Designs: A Survey, Insights, and Perspectives," *IEEE Transactions on Neural Networks and Learning Systems* 35, no. 3 (2024): 2969–2983, https://doi.org/10.1109/TNNLS.2023.3290974.

38. D. Yang, W. Zhang, and Y. Jiang, "Mechanical Fault Diagnosis Based on Deep Transfer Learning: A Review," *Measurement Science and Technology* 34, no. 11 (2023): 112001, https://doi.org/10.1088/1361-6501/ace7e6.

39. S. J. Pan and Q. Yang, "A Survey on Transfer Learning," *IEEE Transactions on Knowledge and Data Engineering* 22, no. 10 (2010): 1345–1359, https://doi.org/10.1109/TKDE.2009.191.

40. X. Chen, R. Yang, Y. Xue, M. Huang, R. Ferrero, and Z. Wang, "Deep Transfer Learning for Bearing Fault Diagnosis: A Systematic Review Since 2016," *IEEE Transactions on Instrumentation and Measurement* 72 (2023): 1–21, https://doi.org/10.1109/TIM.2023.3244237.

41. L. Wan, Y. Li, K. Chen, K. Gong, and C. Li, "A Novel Deep Convolution Multi-Adversarial Domain Adaptation Model for Rolling Bearing Fault Diagnosis," *Measurement* 191 (2022): 110752, https://doi.org/10.1016/j.measurement.2022.110752.

42. B. Zhao, X. Zhang, H. Li, and Z. Yang, "Intelligent Fault Diagnosis of Rolling Bearings Based on Normalized CNN Considering Data Imbalance and Variable Working Conditions," *Knowledge-Based Systems* 199 (2020): 105971, https://doi.org/10.1016/j.knosys.2020.105971.

43. S. Kumar and B. Janet, "DTMIC: Deep Transfer Learning for Malware Image Classification," *Journal of Information Security and Applications* 64 (2022): 103063, https://doi.org/10.1016/j.jisa.2021.103063.

44. J. Yu, Y. Yan, and Z. Z. Lan, "Deep Transfer Learning Based Text Classification Method for Product Reviews," in *Proceedings—2023 16th International Symposium on Computational Intelligence and Design, ISCID 2023* (Institute of Electrical and Electronics Engineers Inc., 2023), 80–83, https://doi.org/10.1109/ISCID59865.2023.00027.

45. A. Haque and K. S. Rao, "Speech Emotion Recognition With Transfer Learning and Multi-Condition Training for Noisy Environments," *International Journal of Speech Technology* 27, no. 2 (2024): 353–365, https://doi.org/10.1007/s10772-024-10109-5.

46. S. Asutkar and S. Tallur, "Deep Transfer Learning Strategy for Efficient Domain Generalisation in Machine Fault Diagnosis," *Scientific Reports* 13, no. 1 (2023): 1–9, https://doi.org/10.1038/s41598-023-33887-5.

47. T. Wang, J. Huan, and M. Zhu, "Instance-Based Deep Transfer Learning," in *Proceedings—2019 IEEE Winter Conference on Applications of Computer Vision, WACV 2019* (Institute of Electrical and Electronics Engineers Inc, 2019), 367–375, https://doi.org/10.1109/WACV.2019.00045.

48. F. Shen, C. Chen, R. Yan, and R. X. Gao, "Bearing Fault Diagnosis Based on SVD Feature Extraction and Transfer Learning Classification," in *Proc. 2015 Progn. Syst. Heal. Manag. Conf. PHM 2015*, vol. 1 (IEEE, 2016), https://doi.org/10.1109/PHM.2015.7380088.

49. D. Xiao, Y. Huang, C. Qin, Z. Liu, Y. Li, and C. Liu, "Transfer Learning With Convolutional Neural Networks for Small Sample Size Problem in Machinery Fault Diagnosis," *Proceedings of the Institution of Mechanical*

Engineers, Part C, Journal of Mechanical Engineering Science 233, no. 14 (2019): 5131–5143, https://doi.org/10.1177/0954406219840381.

50. Y. Miao, Y. Jiang, J. Huang, X. Zhang, and L. Han, "Application of Fault Diagnosis of Seawater Hydraulic Pump Based on Transfer Learning," *Shock and Vibration* 2020, no. 1 (2020): 9630986, https://doi.org/10.1155/2020/9630986.

51. F. Dong, J. Yang, Y. Cai, and L. Xie, "Transfer Learning-Based Fault Diagnosis Method for Marine Turbochargers," *Actuators* 12, no. 4 (2023): 146, https://doi.org/10.3390/act12040146.

52. Z. Du, B. Yang, Y. Lei, X. Li, and N. Li, "A Hybrid Transfer Learning Method for Fault Diagnosis of Machinery Under Variable Operating Conditions," in *2019 Prognostics and System Health Management Conference, PHM-Qingdao 2019* (Institute of Electrical and Electronics Engineers Inc, 2019), https://doi.org/10.1109/PHM-Qingdao46334.2019.8942974.

53. W. Chen, Y. Qiu, Y. Feng, Y. Li, and A. Kusiak, "Diagnosis of Wind Turbine Faults With Transfer Learning Algorithms," *Renewable Energy* 163 (2021): 2053–2067, https://doi.org/10.1016/j.renene.2020.10.121.

54. N. Rezazadeh, F. Annaz, W. A. Jabbar, J. Vieira Filho, and M. De Oliveira, "A Transfer Learning Approach for Mitigating Temperature Effects on Wind Turbine Blades Damage Diagnosis," *Structural Health Monitoring* (2025): 1–23, https://doi.org/10.1177/14759217241313350.

55. Y. Li, Y. Dong, M. Xu, P. Liu, and R. Wang, "Instance Weighting-Based Partial Domain Adaptation for Intelligent Fault Diagnosis of Rotating Machinery," *IEEE Transactions on Instrumentation and Measurement* 72 (2023): 1–14, https://doi.org/10.1109/TIM.2023.3276027.

56. K. Lee, S. Han, V. H. Pham, et al., "Multi-Objective Instance Weighting-Based Deep Transfer Learning Network for Intelligent Fault Diagnosis," *Applied Sciences* 11, no. 5 (2021): 1–21, https://doi.org/10.3390/app11052370.

57. F. Jamil, T. Verstraeten, A. Nowé, C. Peeters, and J. Helsen, "A Deep Boosted Transfer Learning Method for Wind Turbine Gearbox Fault Detection," *Renewable Energy* 197 (2022): 331–341, https://doi.org/10.1016/j.renene.2022.07.117.

58. Y. Shi, A. Deng, M. Deng, et al., "Instance Adaptive Multisource Transfer for Fault Diagnosis of Rotating Machinery Under Variable Working Conditions," *Measurement* 202 (2022): 111797, https://doi.org/10.1016/j.measurement.2022.111797.

59. M. Skowron, "Transfer Learning-Based Fault Detection System of Permanent Magnet Synchronous Motors," *IEEE Access* 12 (2024): 135372–135389, https://doi.org/10.1109/ACCESS.2024.3463970.

60. J. Zhang, Y. Huang, R. Liu, and Z. Wu, "Cross-Domain Decision Method Based on Instance Transfer and Model Transfer for Fault Diagnosis," *Advances in Mechanical Engineering* 16, no. 4 (2024): 1–18, https://doi.org/10.1177/16878132241245836.

61. F. Zhuang, Z. Qi, K. Duan, et al., "A Comprehensive Survey on Transfer Learning," in *Proceedings of the IEEE*, vol. 109 (Institute of Electrical and Electronics Engineers Inc, 2021), 43–76, https://doi.org/10.1109/JPROC.2020.3004555.

62. E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep Domain Confusion: Maximizing for Domain Invariance," 2014, https://arxiv.org/abs/1412.3474v1.

63. W. Dai, G. R. Xue, Q. Yang, and Y. Yu, "Co-Clustering Based Classification for Out-of-Domain Documents," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (ACM, 2007), 210–219, https://doi.org/10.1145/1281192.1281218.

64. J. Shen, Y. Qu, W. Zhang, and Y. Yu, "Wasserstein Distance Guided Representation Learning for Domain Adaptation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32 (AAAI Press, 2018), 4058–4065, https://doi.org/10.1609/aaai.v32i1.11784.

65. C. Chen, Z. Li, J. Yang, and B. Liang, "A Cross Domain Feature Extraction Method Based on Transfer Component Analysis for Rolling Bearing Fault Diagnosis," in *Proceedings of the 29th Chinese Control and Decision Conference, CCDC 2017* (Institute of Electrical and Electronics Engineers Inc, 2017), 5622–5626, https://doi.org/10.1109/CCDC.2017.7978168.

66. W. Xu, Y. Wan, T. Y. Zuo, and X. M. Sha, "Transfer Learning Based Data Feature Transfer for Fault Diagnosis," *IEEE Access* 8 (2020): 76120–76129, https://doi.org/10.1109/ACCESS.2020.2989510.

67. Z. Tong, W. Li, B. Zhang, F. Jiang, and G. Zhou, "Bearing Fault Diagnosis Under Variable Working Conditions Based on Domain Adaptation Using Feature Transfer Learning," *IEEE Access* 6 (2018): 76187–76197, https://doi.org/10.1109/ACCESS.2018.2883078.

68. B. Zhang, W. Li, X. L. Li, and S. K. Ng, "Intelligent Fault Diagnosis Under Varying Working Conditions Based on Domain Adaptive Convolutional Neural Networks," *IEEE Access* 6 (2018): 66367–66384, https://doi.org/10.1109/ACCESS.2018.2878491.

69. B. Yang, Y. Lei, F. Jia, and S. Xing, "An Intelligent Fault Diagnosis Approach Based on Transfer Learning From Laboratory Bearings to Locomotive Bearings," *Mechanical Systems and Signal Processing* 122 (2019): 692–706, https://doi.org/10.1016/j.ymssp.2018.12.051.

70. X. Li, W. Zhang, and Q. Ding, "A Robust Intelligent Fault Diagnosis Method for Rolling Element Bearings Based on Deep Distance Metric Learning," *Neurocomputing* 310 (2018): 77–95, https://doi.org/10.1016/j.neucom.2018.05.021.

71. J. Zhu, N. Chen, and C. Shen, "A New Deep Transfer Learning Method for Bearing Fault Diagnosis Under Different Working Conditions," *IEEE Sensors Journal* 20, no. 15 (2020): 8394–8402, https://doi.org/10.1109/JSEN.2019.2936932.

72. D. Xiao, Y. Huang, L. Zhao, C. Qin, H. Shi, and C. Liu, "Domain Adaptive Motor Fault Diagnosis Using Deep Transfer Learning," *IEEE Access* 7 (2019): 80937–80949, https://doi.org/10.1109/ACCESS.2019.2921480.

73. K. Xu, S. Li, J. Wang, Z. An, W. Qian, and H. Ma, "A Novel Convolutional Transfer Feature Discrimination Network for Unbalanced Fault Diagnosis Under Variable Rotational Speeds," *Measurement Science and Technology* 30, no. 10 (2019): 105107, https://doi.org/10.1088/1361-6501/ab230b.

74. S. Yu, L. Song, S. Pang, M. Wang, X. He, and P. Xie, "M-Net: A Novel Unsupervised Domain Adaptation Framework Based on Multi-Kernel Maximum Mean Discrepancy for Fault Diagnosis of Rotating Machinery," *Complex & Intelligent Systems* 10, no. 3 (2024): 3259–3272, https://doi.org/10.1007/s40747-023-01320-z.

75. M. Huang, J. Yin, S. Yan, and P. Xue, "A Fault Diagnosis Method of Bearings Based on Deep Transfer Learning," *Simulation Modelling Practice and Theory* 122 (2023): 102659, https://doi.org/10.1016/j.simpat.2022.102659.

76. Q. Qian, Y. Wang, T. Zhang, and Y. Qin, "Maximum Mean Square Discrepancy: A New Discrepancy Representation Metric for Mechanical Fault Transfer Diagnosis," *Knowledge-Based Systems* 276 (2023): 110748, https://doi.org/10.1016/j.knosys.2023.110748.

77. J. Li, Z. Ye, J. Gao, Z. Meng, K. Tong, and S. Yu, "Fault Transfer Diagnosis of Rolling Bearings Across Different Devices via Multi-Domain Information Fusion and Multi-Kernel Maximum Mean Discrepancy," *Applied Soft Computing* 159 (2024): 111620, https://doi.org/10.1016/j.asoc.2024.111620.

78. B. Luo and C. Chen, "Gearbox Cross-Speed Fault Diagnosis Based on Transfer Learning of Stator Current," in *2024 3rd International Conference on Robotics, Artificial Intelligence and Intelligent Control, RAIIC 2024* (Institute of Electrical and Electronics Engineers Inc, 2024), 68–72, https://doi.org/10.1109/RAIIC61787.2024.10670727.

79. N. D. Thuan, "Robust Knowledge Transfer for Bearing Diagnosis in Neural Network Models Using Multilayer Maximum Mean Discrepancy Loss Function," *Measurement Science and Technology* 35, no. 12 (2024): 126129, https://doi.org/10.1088/1361-6501/ad7a1b.

80. S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain Adaptation via Transfer Component Analysis," *IEEE Transactions on Neural Networks* 22, no. 2 (2011): 199–210, https://doi.org/10.1109/TNN.2010.2091281.

81. M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer Feature Learning With Joint Distribution Adaptation," in *Proc. IEEE Int. Conf. Comput. Vis* (IEEE, 2013), 2200–2207, https://doi.org/10.1109/ICCV.2013.274.

82. M. Long, Y. Cao, J. Wang, and M. I. Jordan, "Learning Transferable Features With Deep Adaptation Networks," in *32nd Int. Conf. Mach. Learn. ICML*, vol. 1 (IEEE, 2015), 97–105.

83. Z. Xu, D. Huang, T. Min, and Y. Ou, "A Fault Diagnosis Method of Rolling Bearing Integrated With Cooperative Energy Feature Extraction and Improved Least-Squares Support Vector Machine," *Mathematical Problems in Engineering* 2020, no. 1 (2020): 6643167, https://doi.org/10.1155/2020/6643167.

84. D. Li and M. Ma, "A Bearing Fault Diagnosis Method Based on Improved Transfer Component Analysis and Deep Belief Network," *Applied Sciences* 14, no. 5 (2024): 1973, https://doi.org/10.3390/app14051973.

85. R. van de Sand, S. Corasaniti, and J. Reiff-Stephan, "Data-Driven Fault Diagnosis for Heterogeneous Chillers Using Domain Adaptation Techniques," *Control Engineering Practice* 112 (2021): 104815, https://doi.org/10.1016/j.conengprac.2021.104815.

86. Z. Xu, H. Darong, G. Sun, and W. Yongchao, "A Fault Diagnosis Method Based on Improved Adaptive Filtering and Joint Distribution Adaptation," *IEEE Access* 8 (2020): 159683–159695, https://doi.org/10.1109/ACCESS.2020.3020906.

87. T. Han, C. Liu, W. Yang, and D. Jiang, "Deep Transfer Network With Joint Distribution Adaptation: A New Intelligent Fault Diagnosis Framework for Industry Application," *ISA Transactions* 97 (2020): 269–281, https://doi.org/10.1016/j.isatra.2019.08.012.

88. Y. Qin, Q. Qian, J. Luo, and H. Pu, "Deep Joint Distribution Alignment: A Novel Enhanced-Domain Adaptation Mechanism for Fault Transfer Diagnosis," *IEEE Transactions on Cybernetics* 53, no. 5 (2023): 3128–3138, https://doi.org/10.1109/TCYB.2022.3162957.

89. Q. Chang, C. Fang, L. Fan, W. Zhou, and X. Meng, "A Dynamic Weighted Joint Distribution Domain Adaptation Network for Cross-Machine Fault Diagnosis of Rolling Bearings," *Structural Health Monitoring* (2025), https://doi.org/10.1177/14759217241312080.

90. J. Xiong, S. Cui, and H. Tang, "A Novel Intelligent Bearing Fault Diagnosis Method Based on Signal Process and Multi-Kernel Joint Distribution Adaptation," *Scientific Reports* 13, no. 1 (2023): 1–16, https://doi.org/10.1038/s41598-023-31648-y.

91. C. Liu and F. Dong, "A New Framework Based on Supervised Joint Distribution Adaptation for Bearing Fault Diagnosis Across Diverse Working Conditions," *Shock and Vibration* 2024, no. 1 (2024): 8296809, https://doi.org/10.1155/2024/8296809.

92. H. Xu, X. Peng, J. Wang, J. Liu, and C. He, "Adaptive Graph-Guided Joint Soft Clustering and Distribution Alignment for Cross-Load and Cross-Device Rotating Machinery Fault Transfer Diagnosis," *Measurement Science and Technology* 35, no. 4 (2024): 045009, https://doi.org/10.1088/1361-6501/ad1ba3.

93. Y. Geng, G. Tang, and H. Wang, "Domain Adaptation With Joint Distribution Alignment Adversarial Learning for Open-Set Bearing Intelligent Fault Diagnosis," *IEEE Sensors Journal* 25, no. 14 (2025): 26507–26519, https://doi.org/10.1109/JSEN.2025.3576833.

94. Z. Wan, R. Yang, and M. Huang, "Deep Transfer Learning-Based Fault Diagnosis for Gearbox Under Complex Working Conditions," *Shock and Vibration* 2020, no. 1 (2020): 8884179, https://doi.org/10.1155/2020/8884179.

95. C. Che, H. Wang, X. Ni, and Q. Fu, "Domain Adaptive Deep Belief Network for Rolling Bearing Fault Diagnosis," *Computers and Industrial Engineering* 143 (2020): 106427, https://doi.org/10.1016/j.cie.2020.106427.

96. H. Xu, J. Liu, X. Peng, J. Wang, and C. He, "Deep Dynamic Adaptation Network: A Deep Transfer Learning Framework for Rolling Bearing Fault Diagnosis Under Variable Working Conditions," *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 45, no. 1 (2023): 1–17, https://doi.org/10.1007/s40430-022-03950-9.

97. X. Lu, L. Song, C. Han, Q. Jiang, W. Xu, and H. Wang, "Semi-Supervised Contrastive Domain Adaptation Network for Fault Diagnosis of Rotating Machinery Under Cross-Working Conditions," *IEEE Internet of Things Journal* 12 (2025): 21552–21564, https://doi.org/10.1109/JIOT.2025.3547406.

98. W. Qian, S. Li, J. Wang, Y. Xin, and H. Ma, "A New Deep Transfer Learning Network for Fault Diagnosis of Rotating Machine Under Variable Working Conditions," in *Proceedings—2018 Prognostics and System Health Management Conference, PHM-Chongqing 2018* (Institute of Electrical and Electronics Engineers Inc, 2019), 1010–1016, https://doi.org/10.1109/PHM-Chongqing.2018.00180.

99. X. Li, T. Yu, D. Li, et al., "A Migration Learning Method Based on Adaptive Batch Normalization Improved Rotating Machinery Fault Diagnosis," *Sustainability* 15, no. 10 (2023): 8034, https://doi.org/10.3390/su15108034.

100. X. Li, K. Su, D. Li, Q. He, Z. Xie, and X. Kong, "Transfer Learning for Bearing Fault Diagnosis: Adaptive Batch Normalization and Combined Optimization Method," *Measurement Science and Technology* 35, no. 4 (2024): 046106, https://doi.org/10.1088/1361-6501/ad19c2.

101. X. Li, P. Yuan, X. Wang, D. Li, Z. Xie, and X. Kong, "An Unsupervised Transfer Learning Bearing Fault Diagnosis Method Based on Depthwise Separable Convolution," *Measurement Science and Technology* 34, no. 9 (2023): 095401, https://doi.org/10.1088/1361-6501/acda55.

102. C. Cheng, B. Zhou, G. Ma, D. Wu, and Y. Yuan, "Wasserstein Distance Based Deep Adversarial Transfer Learning for Intelligent Fault Diagnosis With Unlabeled or Insufficient Labeled Data," *Neurocomputing* 409 (2020): 35–45, https://doi.org/10.1016/j.neucom.2020.05.040.

103. Z. Zhu, L. Wang, G. Peng, and S. Li, "WDA: An Improved Wasserstein Distance-Based Transfer Learning Fault Diagnosis Method," *Sensors (Basel)* 21, no. 13 (2021): 4394, https://doi.org/10.3390/s21134394.

104. N. Lei, J. Cui, J. Han, X. Chen, and Y. Tang, "Rolling Bearing Fault Diagnosis Using Deep Transfer Learning Based on Joint Generalized Sliced Wasserstein Distance," *IEEE Access* 12 (2024): 41452–41463, https://doi.org/10.1109/ACCESS.2024.3375400.

105. M. Hemmer, H. Van Khang, K. G. Robbersmyr, T. I. Waag, and T. J. J. Meyer, "Fault Classification of Axial and Radial Roller Bearings Using Transfer Learning Through a Pretrained Convolutional Neural Network," *Designs* 2, no. 4 (2018): 1–17, https://doi.org/10.3390/designs2040056.

106. P. Cao, S. Zhang, and J. Tang, "Preprocessing-Free Gear Fault Diagnosis Using Small Datasets With Deep Convolutional Neural Network-Based Transfer Learning," *IEEE Access* 6 (2018): 26241–26253, https://doi.org/10.1109/ACCESS.2018.2837621.

107. J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding," in *NAACL HLT 2019–2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies – Proceedings of the Conference* (Association for Computational Linguistics, 2019), 4171–4186, https://doi.org/10.18653/V1/N19-1423.

108. T. B. Brown, B. Mann, N. Ryder, et al., "Language Models Are Few-Shot Learners," in *Advances in Neural Information Processing Systems* (ACM, 2020), 1877–1901, https://commoncrawl.org/the-data/.

109. E. Kotei and R. Thirunavukarasu, "A Systematic Review of Transformer-Based Pre-Trained Language Models Through Self-Supervised Learning," *Information (Basel)* 14, no. 3 (2023): 187, https://doi.org/10.3390/info14030187.

110. P. N. Srinivasu, G. J. Lakshmi, S. C. Narahari, J. Shafi, J. Choi, and M. F. Ijaz, "Enhancing Medical Image Classification via Federated Learning and Pre-Trained Model," *Egyptian Informatics Journal* 27 (2024): 100530, https://doi.org/10.1016/j.eij.2024.100530.

111. J. Niu, J. Pan, Z. Qin, F. Huang, and H. Qin, "Small-Sample Bearings Fault Diagnosis Based on ResNet18 With Pre-Trained and Fine-Tuned Method," *Applied Sciences* 14, no. 12 (2024): 5360, https://doi.org/10.3390/app14125360.

112. Y. Cheng, S. Shao, T. Niu, Y. Zhao, and K. Jiao, "Intelligent Fault Diagnosis of Gearbox Based on Residual Network and Data Enhancement," in *2024 4th Int. Signal Process. Commun. Eng. Manag. Conf* (IEEE, 2024), 1054–1061, https://doi.org/10.1109/ISPCEM64498.2024.00186.

113. C. V. Prasshanth, S. Naveen Venkatesh, T. K. Mahanta, N. R. Sakthivel, and V. Sugumaran, "Fault Diagnosis of Monoblock Centrifugal Pumps Using Pre-Trained Deep Learning Models and Scalogram Images," *Engineering Applications of Artificial Intelligence* 136 (2024): 109022, https://doi.org/10.1016/j.engappai.2024.109022.

114. S. S. Zhong, S. Fu, and L. Lin, "A Novel Gas Turbine Fault Diagnosis Method Based on Transfer Learning With CNN," *Measurement: Journal of the International Measurement Confederation* 137 (2019): 435–453, https://doi.org/10.1016/j.measurement.2019.01.022.

115. P. Ma, H. Zhang, W. Fan, C. Wang, G. Wen, and X. Zhang, "A Novel Bearing Fault Diagnosis Method Based on 2D Image Representation and Transfer Learning-Convolutional Neural Network," *Measurement Science and Technology* 30, no. 5 (2019): 055402, https://doi.org/10.1088/1361-6501/ab0793.

116. J. Zhou, X. Yang, L. Zhang, S. Shao, and G. Bian, "Multisignal VGG19 Network With Transposed Convolution for Rotating Machinery Fault Diagnosis Based on Deep Transfer Learning," *Shock and Vibration* 2020 (2020): 1–12, https://doi.org/10.1155/2020/8863388.

117. Z. Chen, J. Cen, and J. Xiong, "Rolling Bearing Fault Diagnosis Using Time-Frequency Analysis and Deep Transfer Convolutional Neural Network," *IEEE Access* 8 (2020): 150248–150261, https://doi.org/10.1109/ACCESS.2020.3016888.

118. W. Zhang, T. Zhang, G. Cui, and Y. Pan, "Intelligent Machine Fault Diagnosis Using Convolutional Neural Networks and Transfer Learning," *IEEE Access* 10 (2022): 50959–50973, https://doi.org/10.1109/ACCESS.2022.3173444.

119. Y. Liu, J. Li, Q. Li, and Q. Wang, "Transfer Learning With Inception ResNet-Based Model for Rolling Bearing Fault Diagnosis," *Journal of Advanced Mechanical Design, Systems, and Manufacturing* 16, no. 2 (2022): 1–19, https://doi.org/10.1299/jamdsm.2022jamdsm0023.

120. Y. Chen, H. Shi, Y. Jin, and Y. Liu, "Study on Fault Diagnosis of Ultra-Low-Speed Bearings Under Variable Working Conditions Based on Improved EfficientNet Network," *Insight – Non-Destructive Testing and Condition Monitoring* 66, no. 2 (2024): 94–102, https://doi.org/10.1784/insi.2024.66.2.94.

121. J. Wang, Z. Mo, H. Zhang, and Q. Miao, "A Deep Learning Method for Bearing Fault Diagnosis Based on Time-Frequency Image," *IEEE Access* 7 (2019): 42373–42383, https://doi.org/10.1109/ACCESS.2019.2907131.

122. A. Jha, S. Singh, and V. Sharma, "Wavelet Based Bearing Fault Diagnosis Using Transfer Learning," in *2024 International Conference on Computer, Electronics, Electrical Engineering and Their Applications, IC2E3 2024* (Institute of Electrical and Electronics Engineers Inc, 2024), https://doi.org/10.1109/IC2E362166.2024.10827208.

123. S. Shao, S. McAleer, R. Yan, and P. Baldi, "Highly Accurate Machine Fault Diagnosis Using Deep Transfer Learning," *IEEE Transactions on Industrial Informatics* 15, no. 4 (2019): 2446–2455, https://doi.org/10.1109/TII.2018.2864759.

124. L. Wen, X. Li, X. Li, and L. Gao, "A New Transfer Learning Based on VGG-19 Network for Fault Diagnosis," in *Proceedings of the 2019 IEEE 23rd International Conference on Computer Supported Cooperative Work in Design, CSCWD 2019* (IEEE, 2019), 205–209, https://doi.org/10.1109/CSCWD.2019.8791884.

125. J. Su and H. Wang, "Fine-Tuning and Efficient VGG16 Transfer Learning Fault Diagnosis Method for Rolling Bearing," in *Mechanisms and Machine Science* (Springer, 2023), 453–461, https://doi.org/10.1007/978-3-030-99075-6_37.

126. A. Ibrahim, F. Anayi, and M. Packianather, "New Transfer Learning Approach Based on a CNN for Fault Diagnosis †," *Engineering Proceedings* 24, no. 1 (2022): 16, https://doi.org/10.3390/IECMA2022-12905.

127. D. Saha, M. E. Hoque, and M. E. H. Chowdhury, "Enhancing Bearing Fault Diagnosis Using Transfer Learning and Random Forest Classification: A Comparative Study on Variable Working Conditions," *IEEE Access* 12 (2024): 5986–6000, https://doi.org/10.1109/ACCESS.2023.3347345.

128. L. Wen, X. Li, and L. Gao, "A Transfer Convolutional Neural Network for Fault Diagnosis Based on ResNet-50," *Neural Computing and Applications* 32, no. 10 (2020): 6111–6124, https://doi.org/10.1007/s00521-019-04097-w.

129. P. Kumar, P. Kumar, A. S. Hati, and H. S. Kim, "Deep Transfer Learning Framework for Bearing Fault Detection in Motors," *Mathematics* 10, no. 24 (2022): 4683, https://doi.org/10.3390/math10244683.

130. C. He, Y. Cao, Y. Yang, Y. Liu, X. Liu, and Z. Cao, "Fault Diagnosis of Rotating Machinery Based on the Improved Multidimensional Normalization ResNet," *IEEE Transactions on Instrumentation and Measurement* 72 (2023): 1–11, https://doi.org/10.1109/TIM.2023.3293554.

131. X. Yu, S. Wang, H. Xu, et al., "Intelligent Fault Diagnosis of Rotating Machinery Under Variable Working Conditions Based on Deep Transfer Learning With Fusion of Local and Global Time–Frequency Features," *Structural Health Monitoring* 23, no. 4 (2024): 2238–2254, https://doi.org/10.1177/14759217231199427.

132. S. Liu, Z. Ji, Z. Zhang, and Y. Wang, "An Improved Deep Transfer Learning Method for Rotating Machinery Fault Diagnosis Based on Time Frequency Diagram and Pretraining Model," *IEEE Transactions on Instrumentation and Measurement* 73 (2024): 1–12, https://doi.org/10.1109/TIM.2023.3345907.

133. X. Zhang, X. Zhang, W. Liang, and F. He, "Research on Rolling Bearing Fault Diagnosis Based on Parallel Depthwise Separable ResNet Neural Network With Attention Mechanism," *Expert Systems with Applications* 286 (2025): 128105, https://doi.org/10.1016/j.eswa.2025.128105.

134. V. Jaiswal, T. Narendiranath Babu, P. Murugan, and D. Rama Prabha, "Fault Analysis on Deep Groove Ball Bearing Using ResNet50 and AlexNet50 Algorithms," *Scientific Reports* 15, no. 1 (2025): 1–22, https://doi.org/10.1038/s41598-025-97410-8.

135. H. Ziad, A. Al-Dujaili, and A. J. Humaidi, "Electrical Faults Classification in Permanent Magnet Synchronous Motor Using ResNet Neural Network," *International Review of Applied Sciences and Engineering* 15, no. 3 (2024): 355–364, https://doi.org/10.1556/1848.2024.00789.

136. H. Jung, S. Choi, and B. Lee, "Rotor Fault Diagnosis Method Using CNN-Based Transfer Learning With 2D Sound Spectrogram Analysis," *Electronics* 12, no. 3 (2023): 480, https://doi.org/10.3390/electronics12030480.

137. L. Xu, S. S. Teoh, and H. Ibrahim, "A Deep Learning Approach for Electric Motor Fault Diagnosis Based on Modified InceptionV3," *Scientific Reports* 14, no. 1 (2024): 1–15, https://doi.org/10.1038/s41598-024-63086-9.

138. L. Wei, X. Peng, and Y. Cao, "Enhanced Fault Diagnosis of Rolling Bearings Using an Improved Inception-Lstm Network," *Nondestructive*

Testing and Evaluation 40, no. 7 (2025): 3274–3293, https://doi.org/10.1080/10589759.2024.2402549.

139. Z. Li, P. Wang, Z. Yang, X. Li, and R. Jia, "IInception-CBAM-IBiGRU Based Fault Diagnosis Method for Asynchronous Motors," *Scientific Reports* 14, no. 1 (2024): 1–16, https://doi.org/10.1038/s41598-024-55367-0.

140. Y. Li, Y. Gu, C. Shi, and T. Li, "Fault Diagnosis Method of Planetary Gearbox Based on Transfer Learning and EfficientNet," in *2023 Global Reliability and Prognostics and Health Management Conference, PHM-Hangzhou 2023* (Institute of Electrical and Electronics Engineers Inc, 2023), https://doi.org/10.1109/PHM-HANGZHOU58797.2023.10482588.

141. B. Hu, J. Tang, J. Wu, and J. Qing, "An Attention EfficientNet-Based Strategy for Bearing Fault Diagnosis Under Strong Noise," *Sensors (Basel)* 22, no. 17 (2022): 6570, https://doi.org/10.3390/s22176570.

142. T. Han, C. Liu, W. Yang, and D. Jiang, "Learning Transferable Features in Deep Convolutional Neural Networks for Diagnosing Unseen Machine Conditions," *ISA Transactions* 93 (2019): 341–353, https://doi.org/10.1016/j.isatra.2019.03.017.

143. Z. Chen, K. Gryllias, and W. Li, "Intelligent Fault Diagnosis for Rotary Machinery Using Transferable Convolutional Neural Network," *IEEE Transactions on Industrial Informatics* 16, no. 1 (2020): 339–349, https://doi.org/10.1109/TII.2019.2917233.

144. M. J. Hasan, M. M. M. Islam, and J. M. Kim, "Acoustic Spectral Imaging and Transfer Learning for Reliable Bearing Fault Diagnosis Under Variable Speed Conditions," *Measurement: Journal of the International Measurement Confederation* 138 (2019): 620–631, https://doi.org/10.1016/j.measurement.2019.02.075.

145. Z. Xing, Y. Liu, Q. Wang, and J. Fu, "Fault Diagnosis of Rotating Parts Integrating Transfer Learning and ConvNeXt Model," *Scientific Reports* 15, no. 1 (2025): 1–18, https://doi.org/10.1038/s41598-024-84783-5.

146. P. Aduwenye, C. Nutakor, L. Roininen, and J. Sopanen, "Parameter-Based Transfer Learning for Bearing Fault Diagnosis Using Small Samples Under Variable Working Conditions," *IEEE Access* 13 (2025): 44214–44230, https://doi.org/10.1109/ACCESS.2025.3548549.

147. K. Zhao, H. Jiang, X. Li, and R. Wang, "Ensemble Adaptive Convolutional Neural Networks With Parameter Transfer for Rotating Machinery Fault Diagnosis," *International Journal of Machine Learning and Cybernetics* 12, no. 5 (2021): 1483–1499, https://doi.org/10.1007/s13042-020-01249-6.

148. I. Goodfellow, J. Pouget-Abadie, M. Mirza, et al., "Generative Adversarial Networks," *Communications of the ACM* 63, no. 11 (2020): 139–144, https://doi.org/10.1145/3422622.

149. S. Karpakam and N. Kumareshan, "Enhanced Brain Tumor Detection and Classification Using a Deep Image Recognition Generative Adversarial Network (DIR-GAN): A Comparative Study on MRI, X-Ray, and FigShare Datasets," *Neural Computing and Applications* 37, no. 14 (2025): 8731–8758, https://doi.org/10.1007/s00521-025-11031-w.

150. R. MeenaPrakash, B. Kamali, M. Vimala, K. Madhuvandhana, and P. Krishnaleela, "A DenseNet-Enhanced GAN Model for Classification of Medical Images Into Original and Fake," in *2025 Int. Conf. Multi-Agent Syst. Collab. Intell* (IEEE, 2025), 1540–1545, https://doi.org/10.1109/ICMSCI62561.2025.10894026.

151. Y. Zou, W. Zhao, T. Liu, X. Zhang, and Y. Shi, "Research on High-Speed Train Bearing Fault Diagnosis Method Based on Domain-Adversarial Transfer Learning," *Applied Sciences* 14, no. 19 (2024): 8666, https://doi.org/10.3390/app14198666.

152. S. Han, Z. Feng, Y. Zhang, M. Du, and Y. Yang, "Intelligent Fault Diagnosis of Planetary Gearbox Across Conditions Based on Subdomain Distribution Adversarial Adaptation," *Sensors (Basel)* 24, no. 21 (2024): 7017, https://doi.org/10.3390/s24217017.

153. Z. Dong, H. An, S. Liu, et al., "Domain Adversarial Transfer Fault Diagnosis Method of an Axial Piston Pump Based on a Multi-Scale Attention Mechanism," *Measurement* 239 (2025): 115455, https://doi.org/10.1016/j.measurement.2024.115455.

154. H. Wang, J. Lin, and Z. Zhang, "Single Imbalanced Domain Generalization Network for Intelligent Fault Diagnosis of Compressors in HVAC Systems Under Unseen Working Conditions," *Energy and Buildings* 312 (2024): 114192, https://doi.org/10.1016/j.enbuild.2024.114192.

155. J. Shi, X. Wang, S. Lu, J. Zheng, H. Dong, and J. Zhang, "An Adversarial Multisource Data Subdomain Adaptation Model: A Promising Tool for Fault Diagnosis of Induction Motor Under Cross-Operating Conditions," *IEEE Transactions on Instrumentation and Measurement* 72 (2023): 1–14, https://doi.org/10.1109/TIM.2023.3280493.

156. S. Yao, Q. Kang, M. C. Zhou, M. J. Rawa, and A. Abusorrah, "A Survey of Transfer Learning for Machinery Diagnostics and Prognostics," *Artificial Intelligence Review* 56, no. 4 (2023): 2871–2922, https://doi.org/10.1007/s10462-022-10230-4.

157. Q. Li, C. Shen, L. Chen, and Z. Zhu, "Knowledge Mapping-Based Adversarial Domain Adaptation: A Novel Fault Diagnosis Method With High Generalizability Under Variable Working Conditions," *Mechanical Systems and Signal Processing* 147 (2021): 107095, https://doi.org/10.1016/j.ymssp.2020.107095.

158. Y. Qin, Q. Yao, Y. Wang, and Y. Mao, "Parameter Sharing Adversarial Domain Adaptation Networks for Fault Transfer Diagnosis of Planetary Gearboxes," *Mechanical Systems and Signal Processing* 160 (2021): 107936, https://doi.org/10.1016/j.ymssp.2021.107936.

159. J. Jiao, M. Zhao, J. Lin, and K. Liang, "Residual Joint Adaptation Adversarial Network for Intelligent Transfer Fault Diagnosis," *Mechanical Systems and Signal Processing* 145 (2020): 106962, https://doi.org/10.1016/j.ymssp.2020.106962.

160. J. Jiao, J. Lin, M. Zhao, and K. Liang, "Double-Level Adversarial Domain Adaptation Network for Intelligent Fault Diagnosis," *Knowledge-Based Systems* 205 (2020): 106236, https://doi.org/10.1016/j.knosys.2020.106236.

161. Y. Deng, D. Huang, S. Du, G. Li, C. Zhao, and J. Lv, "A Double-Layer Attention Based Adversarial Network for Partial Transfer Learning in Machinery Fault Diagnosis," *Computers in Industry* 127 (2021): 103399, https://doi.org/10.1016/j.compind.2021.103399.

162. T. Gao, J. Yang, and Q. Tang, "A Multi-Source Domain Information Fusion Network for Rotating Machinery Fault Diagnosis Under Variable Operating Conditions," *Information Fusion* 106 (2024): 102278, https://doi.org/10.1016/j.inffus.2024.102278.

163. T. Gao, J. Yang, W. Wang, and X. Fan, "A Domain Feature Decoupling Network for Rotating Machinery Fault Diagnosis Under Unseen Operating Conditions," *Reliability Engineering and System Safety* 252 (2024): 110449, https://doi.org/10.1016/j.ress.2024.110449.

164. Y. Xu, J. Liu, Z. Wan, D. Zhang, and D. Jiang, "Rotor Fault Diagnosis Using Domain-Adversarial Neural Network With Time-Frequency Analysis," *Machines* 10, no. 8 (2022): 610, https://doi.org/10.3390/machines10080610.

165. B. Wang, P. Liang, L. Zhang, X. Wang, X. Yuan, and Z. Zhou, "Enhancing Robustness of Cross-Machine Fault Diagnosis via an Improved Domain Adversarial Neural Network and Self-Adversarial Training," *Measurement* 250 (2025): 117113, https://doi.org/10.1016/J.MEASUREMENT.2025.117113.

166. X. He, F. Zhao, N. Song, C. Su, and P. Liu, "A Deep-Transfer-Learning Fault Diagnosis Method for Gearboxes Based on Discriminative Feature Extraction and Improved Domain Adversarial Neural Networks," *Nondestructive Testing and Evaluation* (2025), https://doi.org/10.1080/10589759.2025.2495802.

167. C. Chen, X. Li, J. Shi, et al., "A Triple Domain Adversarial Neural Network for Bearing Fault Diagnosis," *Mechanical Systems and Signal Processing* 238 (2025): 113202, https://doi.org/10.1016/j.ymssp.2025.113202.

168. S. Xiang, J. Zhang, H. Gao, D. Shi, and L. Chen, "A Deep Transfer Learning Method for Bearing Fault Diagnosis Based on Domain Separation and Adversarial Learning," *Shock and Vibration* 2021, no. 1 (2021): 5540084, https://doi.org/10.1155/2021/5540084.

169. J. Shao, Z. Huang, Y. Zhu, J. Zhu, and D. Fang, "Rotating Machinery Fault Diagnosis by Deep Adversarial Transfer Learning Based on Subdomain Adaptation," *Advances in Mechanical Engineering* 13, no. 8 (2021), https://doi.org/10.1177/16878140211040226.

170. J. Tang, X. Qiao, X. Liu, and Z. Zhang, "Domain Adversarial Transfer Learning for Fault Diagnosis of Rotating Parts in Cross Working Condition Injection Molding Machines," in *15th Global Reliability and Prognostics and Health Management Conference, PHM-Beijing 2024* (Institute of Electrical and Electronics Engineers Inc, 2024), 1–11, https://doi.org/10.1109/PHM-BEIJING63284.2024.10874448.

171. S. Liu, H. Wang, J. Tang, and X. Zhang, "Research on Fault Diagnosis of Gas Turbine Rotor Based on Adversarial Discriminative Domain Adaption Transfer Learning," *Measurement* 196 (2022): 111174, https://doi.org/10.1016/j.measurement.2022.111174.

172. X. Chen, H. Shao, Y. Xiao, S. Yan, B. Cai, and B. Liu, "Collaborative Fault Diagnosis of Rotating Machinery via Dual Adversarial Guided Unsupervised Multi-Domain Adaptation Network," *Mechanical Systems and Signal Processing* 198 (2023): 110427, https://doi.org/10.1016/j.ymssp.2023.110427.

173. S. Liu, H. Jiang, Z. Wu, Z. Yi, and R. Wang, "Intelligent Fault Diagnosis of Rotating Machinery Using a Multi-Source Domain Adaptation Network With Adversarial Discrepancy Matching," *Reliability Engineering and System Safety* 231 (2023): 109036, https://doi.org/10.1016/j.ress.2022.109036.

174. Z. Wu, H. Jiang, S. Liu, Y. Liu, and W. Yang, "Conditional Distribution-Guided Adversarial Transfer Learning Network With Multi-Source Domains for Rolling Bearing Fault Diagnosis," *Advanced Engineering Informatics* 56 (2023): 101993, https://doi.org/10.1016/j.aei.2023.101993.

175. X. Wang, H. Jiang, M. Mu, and Y. Dong, "A Trackable Multi-Domain Collaborative Generative Adversarial Network for Rotating Machinery Fault Diagnosis," *Mechanical Systems and Signal Processing* 224 (2025): 111950, https://doi.org/10.1016/j.ymssp.2024.111950.

176. L. Guo, Y. Lei, S. Xing, T. Yan, and N. Li, "Deep Convolutional Transfer Learning Network: A New Method for Intelligent Fault Diagnosis of Machines With Unlabeled Data," *IEEE Transactions on Industrial Electronics* 66, no. 9 (2019): 7316–7325, https://doi.org/10.1109/TIE.2018.2877090.

177. M. Zhang, D. Wang, W. Lu, J. Yang, Z. Li, and B. Liang, "A Deep Transfer Model With Wasserstein Distance Guided Multi-Adversarial Networks for Bearing Fault Diagnosis Under Different Working Conditions," *IEEE Access* 7 (2019): 65303–65318, https://doi.org/10.1109/ACCESS.2019.2916935.

178. X. Wang and F. Liu, "Triplet Loss Guided Adversarial Domain Adaptation for Bearing Fault Diagnosis," *Sensors (Basel)* 20, no. 1 (2020): 320, https://doi.org/10.3390/s20010320.

179. J. Jiao, M. Zhao, and J. Lin, "Unsupervised Adversarial Adaptation Network for Intelligent Fault Diagnosis," *IEEE Transactions on Industrial Electronics* 67, no. 11 (2020): 9904–9913, https://doi.org/10.1109/TIE.2019.2956366.

180. C. Liu, S. Li, H. Chen, X. Xiu, and C. Peng, "Semi-Supervised Joint Adaptation Transfer Network With Conditional Adversarial Learning for Rotary Machine Fault Diagnosis," *Intelligent Robots* 3, no. 2 (2023): 131–143, https://doi.org/10.20517/ir.2023.07.

181. X. Li, T. Yu, X. Wang, D. Li, Z. Xie, and X. Kong, "Fusing Joint Distribution and Adversarial Networks: A New Transfer Learning Method for Intelligent Fault Diagnosis," *Applied Acoustics* 216 (2024): 109767, https://doi.org/10.1016/j.apacoust.2023.109767.

182. Z. Su, W. Jiang, K. Chen, M. Luo, S. Feng, and C. Zhou, "Multi-Adversarial Deep Transfer Network for Multi-Source Open-Set Fault Diagnosis of Rotating Machinery With Category Shift," *Knowledge-Based Systems* 282 (2023): 111106, https://doi.org/10.1016/j.knosys.2023.111106.

183. Z. Wang, H. Xia, W. Yin, and B. Yang, "An Improved Generative Adversarial Network for Fault Diagnosis of Rotating Machine in Nuclear Power Plant," *Annals of Nuclear Energy* 180 (2023): 109434, https://doi.org/10.1016/j.anucene.2022.109434.